

AGRICULTURAL CROP PRODUCTION NC II
COMPETENCY-BASED LEARNING MATERIAL

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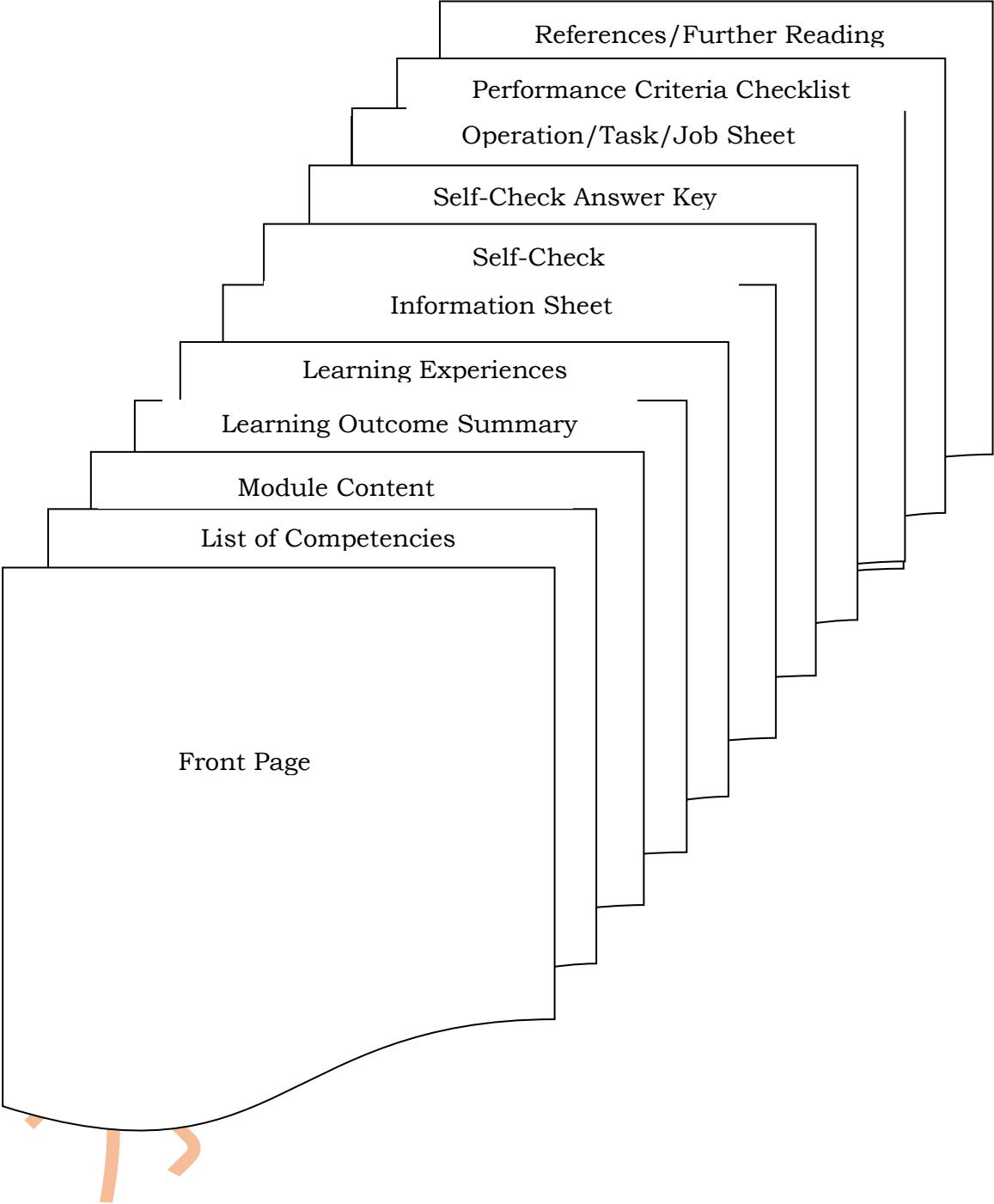
	AGRICULTURAL CROP PRODUCTION NC II		Date Developed: November 16, 2020	
			Issued by:	Page No.
		Developed by: Jumelyn P. Buya	Revision:	



Sector:	Agriculture and Fishery
Qualification Title:	AGRICULTURAL CROP PRODUCTION NC II
Unit of Competency:	PERFORM NURSERY OPERATION
Module Title:	PERFORMING NURSERY OPERATION

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No.
			Revision:	

PARTS OF A COMPETENCY-BASED LEARNING



	AGRICULTURAL CROP PRODUCTION NC II		Date Developed: November 16, 2020	
			Issued by:	Page No.
		Developed by: Jumelyn P. Buya	Revision:	

HOW TO USE THIS COMPETENCY-BASED LEARNING MATERIAL

Welcome!

The unit of competency, “**Perform Nursery Operation**”, is one of the Core Competency of Agricultural Crop Production NC II, a course which comprises the knowledge, skills and attitudes required for National Certificate level II.

The module, **Perform Nursery Operation**, contains training materials and activities related to enhancing the knowledge, skills and attitude of the trainees in agricultural crop for you to complete.

In this module, you are required to go through a series of learning activities in order to complete each learning outcome. In each learning outcome are Information Sheets, Self-Checks, Operation Sheets and Task/Job Sheets. Follow and perform the activities on your own. If you have questions, do not hesitate to ask for assistance from your facilitator.

Remember to:

- Work through all the information and complete the activities in each section.
- Read information sheets and complete the self-check. Suggested references are included to supplement the materials provided in this module.
- Most probably, your trainer will also be your supervisor or manager. He is there to support you and show you the correct way to do things.
- You will be given plenty of opportunities to ask questions and practice on the job. Make sure you practice your new skills during regular work shifts. This way, you will improve your speed, memory and your confidence.
- Use the Self-Checks, Operation Sheets or Task or Job Sheets at the end of each section to test your own progress. Use the Performance Criteria Checklist or procedural Checklist located after the sheet to check your own performance.
- When you feel confident that you have had sufficient practice, ask your Trainer to evaluate you. The results of your assessment will be recorded in your Progress Chart and Accomplishment Chart.

You need to complete this module before you can perform the next module.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

AGRICULTURAL CROP PRODUCTION NC1I

List of Competencies

No.	Unit of Competency	Module Title	Code
1.	Perform nursery operation	Performing nursery operation	AFF 610301
2.	Plant Crops	Planting crops	AFF 610302
3.	Care and maintain crops	Caring and maintaining crops	AFF 610303
4.	Carry-out harvest and post-harvest operations	Carrying-out harvest and post-harvest operations	AFF 610304

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed:	
		November 16, 2020	November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

MODULE CONTENT

UNIT OF COMPETENCY: PERFORM NURSERY OPERATION

CODE: AFF 610301

MODULE TITLE: PERFORMING NURSERY OPERATION

MODULE DESCRIPTOR:

This module covers the knowledge, skills and attitudes required to perform nursery operations for agricultural crops including established nursery shed, preparation and handling of planting materials, preparation of growing media, transplanting germinated seedlings and handling of nursery tools and equipment.

NOMINAL HOURS: 65 HOURS

SUMMARY OF LEARNING OUTCOME

Upon completion of this module, the trainee must be able to:

- 1. Prepare nursery tools, farm implements and simple equipment
- 2. Maintain nursery facilities
- 3. Handle seeds/planting material
- 4. Prepare growing media
- 5. Conduct propagation activities

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

LO 1. Prepare nursery tools, farm implements and simple equipment

Assessment Criteria

- 1. Tools, farm implements and simple equipment are prepared according work requirements.
- 2. Basic pre-operative checking of tools, farm implements and equipment is performed in accordance with manufacturer’s manual and GAP standard.
- 3. Tools with wear and corrosion are segregated and treated according to maintenance plan and procedures.

Contents:

- Use of tools, farm implements and simple equipment
- Segregation and treatment of tools with wear and corrosion
- Good Agricultural Practices (GAP)
- 7’s and 3R

Conditions

- Illustrated materials
- Manuals PPE’s
- Tools

- Digging tools

➤ Propagation tools

➤ Harvesting tools

➤ Measuring tools

GAP

> water pumps

> hand tractor

> plow

> harrow

>sprayer

7’s &

Methodologies:

- Lecture/discussion
- Practical demonstration
- Multimedia viewing

Assessment method

- Written examination

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

LEARNING EXPERIENCES

Learning Outcome 1

LO Title: Prepare Nursery Tools, Farm Implements and simple Equipment

Learning Activities	Special Instructions
1. Read Information Sheet 1.1.-1 on Nursery tools, farm implements and simple equipment's	If you believed you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self- Check 1.1-1	If you get perfect score, proceed to the next activity. If not, read Information Sheet 1.1-1 and answer Self-Check again.
3. Do Task Sheet 1.1.-1	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 1.2-1	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

INFORMATION SHEET 1.1-1

Farm tools, implements and equipment play important role in agricultural crop production. Their availability makes the work much easier and faster. However, familiarization, expertise, skills and knowledge are very much needed in this tool, implements and equipment.

Hand tools- are usually light and are used without the help of animals

Machinery- is a mechanized equipment which can run on electric or diesel or petrol or hydraulic. Machinery is to apply force and control movement for an intended action. Most of farm machinery is still operated by human but there is a gradual introduction of machinery which is smart and can run according to program pre fed into its computer. A common machinery used extensively on farms is TRACTOR.

Examples of Machines:

- 1. Tractor: Probably the most used and most important machinery in a farm. It is used for pulling or pushing agricultural for planting, tilling, ploughing, harrowing etc.
- 2. Harvester: Mechanical harvesting is the order of the day these days due to problems of farm laborers. Larger amount of crop can be harvested quickly and most efficiently.
- 3. Knapsack Sprayer: this machine could be manual, electric or petrol driven. It is used to apply soluble pesticides to plants. A much-used piece of equipment.
- 4. Brush cutter-easier equipment for cutting grasses.

Tractor

Hand tractor

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	



gasoline operated power sprayer
spraying



Electric power sprayer



Heavy duty knapsack sprayer



Manual knapsack sprayer

used during and
irrigation, fertigation
pesticides & foliar

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Sprayers are used to apply soluble or dry chemicals suitable for fighting pests, virus and infections in the plants. There are a number of sprayers distinguished by their design – manual, electric and petrol driven. The nozzles of sprayers also come in different designs to provide for desired droplet size and spread geometry.



soil pH



EC meter

Soil pH meter- use to measure the degree of acidity and alkalinity of the soil



2-stroke gasoline type brush cutter



4-stroke gasoline type

Brush cutter-(also called a brushsaw or clearing saw) is a powered garden or agricultural tool used to trim weeds, small trees, and other foliage not

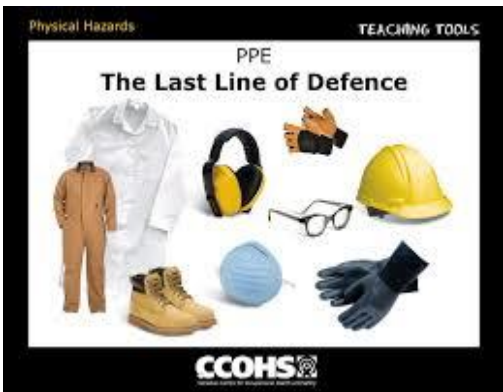
	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

accessible by a lawn **mower** or rotary **mower**. Various blades or **trimmer** heads can be attached to the machine for specific applications.



Wheel borrow- a small cart with a single wheel at the front and two supporting legs and two handles at the rear, used typically for carrying loads in building-work or gardening.

Personal protective equipment (PPE)



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	



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	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:
			Page No.
			Revision:

SELF-CHECK 1.1-1

- Evaluation:
- 1-10 Give some tools, materials and equipment used during nursery operation
 - 11-14 Soil media mixture
 - 15-20 Propagation Techniques



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

SELF-CHECK ANSWER 1.1-1

- 1. Nursery shed
- 2. Seeds
- 3. planting materials
- 4. seedling media
- 5. Seed box
- 6. Shovel
- 7. Wheel borrow
- 8. Power sprayer for irrigation& spraying
- 9. Hand spray
- 10. Seedling trays
- 11. Coco dust
- 12. Black soil
- 13. Peat moss
- 14. Compost
- 15. Grafting layer
- 16. Budding
- 17. Stem cuttings
- 18. Seed
- 19. Runners
- 20. Tissue culture

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

OPERATION/TASK/JOB SHEET 1.1-1

TASK SHEET 1.1.1	
Unit of Competency: Perform nursery operations	
Module Title: Performing nursery operations	
Performance objective: Given the following tools, materials and equipment, you are required to perform nursery operation	
Supplies/Materials: Seedling media-mixture of 1:1:1 part of coco dust, black soil, peat moss, seeds	
Equipment: Power sprayer	
Steps/Procedures 1. Prepare all the materials needed such 3 components of seedling media, seeds, seedling trays, seed box with seedlings ready to prick. 2. Mix all the seedling media of 1:1:1part ratio, maintain moisture and put in the seedling trays. 3. Direct or indirect can be done through pricking 3-5 days seedlings from seedling box to the seedling trays.	
Assessment Method •Written examination •Direct observation •Demonstration	

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Page No.
		Revision:	

PERFORMANCE CRITERIA CHECKLIST 1.1-1

Did I	Yes	No
• Prepare materials, tools and equipment in nursery operation		
•Can do the actual mixing of seedling media according to the ratio of mixture		
•Can do sanitizing of materials and tools prior to usage		
•Can follow the good agricultural practices (GAP) standards in nursery operation		
•Can do actual hands-on in pricking seedlings in the seedling trays		
•Can identify the 3 R's in farm waste management		
•Can apply the 5'S in the workplace		
•Can do direct and indirect seeding		
•Can do hardening of seedlings prior to transplanting		

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

REFERNCES/FURTHER READING

- 1. Training Regulations (TR)- TESDA
- 2. Competency Based Curriculum- TESDA
- 3. Agricultural Crop Production: Beginners, An Introduction
- 4. The Science and Practice of Crop Production-RM Lantican
- 5. Ornamental Horticulture- T.J. Rimando
- 6. Postharvest Technology for Southeast Asian Perishable Crops-O.K. Bautista



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

LO2. Maintain Nursery Facilities

Assessment Criteria

- 1. Nursery sanitation is maintained according to GAP standard.
- 2. Repair and maintenance of nursery facilities are performed to maximized their efficiency and effectiveness.
- 3. Preventive measures are applied for inclement weather.
- 4. Safety measures are practice according to OSHS.

Contents

•Good Agricultural Practices (GAP)

- Nursery management
- Nursery facilities
- Sanitation
- Repair and maintenance of nursery facilities
- 3R's and 7's

Conditions

- Manual
- Illustrated materials/shed/materials
- PPE's

Methodologies

- Lecture/discussion
- Practical demonstration
- Multimedia viewing

Assessment methods

- Written examination
- Demonstration with questioning

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

Learning Experiences

Learning Outcome 2
LO2. Title: Maintain Nursery Facilities

Learning Activities	Special Instructions
1. Read Information Sheet 2.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 2.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 2.1-1 and answer Self- Check 2.1-1 again.
3. Do Task Sheet 2.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 2.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

INFORMATION SHEET 2.1-1

Nursery- a place where plants are propagated and grown to usable size.



Greenhouse- it is a structure with walls and roof made chiefly of transparent material, such as glass, in which plants requiring regulated climatic condition.

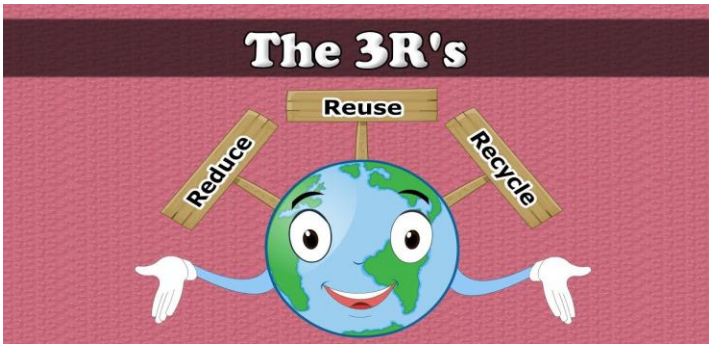


Hygiene- it is a set of practices performed for the preservation of health.

Sustainability: reduce, reuse and recycle

At one point or another we've all heard about the “**three R's**” of sustainability: reduce, reuse and recycle. The goal of the **three R's** is to prevent waste and conserve natural resources. ... It takes resources to manufacture, transport, and dispose of products, so reduction minimizes the use of new resources.

Reduce, Reuse, Recycle. Recycling, while accessible and easy, is not the best option of the **three** for **environmental** health. In fact, of the **three**, it is the least environmentally friendly.



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Page No.
		Revision:	

JAPANESE	ENGLISH	MEANING	TYPICAL EXAMPLE
<u>SEIRI</u>	Sort	Organization	Throw away rubbish
<u>SEITON</u>	Set in Order	Neatness	30-second retrieval of a document
<u>SEISO</u>	Shine	Cleaning	Individual cleaning responsibility
<u>SEIKETSU</u>	Standardize	Standardization	Transparency of storage
<u>SHITSUKE</u>	Sustain	Discipline	Do 5-S daily



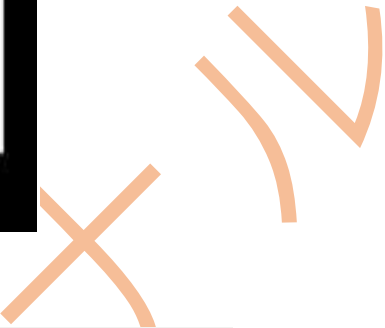
Sanitation refers to public health conditions related to clean drinking water and adequate treatment and disposal of human excreta and sewage. A **sanitation** system includes the capture, storage, transport, treatment and disposal or reuse of human excreta and wastewater.



Personal Protective Equipment (PPE)- commonly referred to as "**PPE**", is **equipment** worn to minimize exposure to hazards that cause serious workplace injuries and illnesses. These injuries and illnesses may result

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:		Date Developed: November 16, 2020	
		November 16, 2020		Issued by:	Page No.
		Developed by: Jumelyn P. Buya			
				Revision:	

from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards.



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Learning Outcome 3

LO3. Title: Handle seeds/ Planting Material

Learning Activities	Special Instructions
1. Read Information Sheet 3.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 3.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 2.1-1 and answer Self- Check 2.1-1 again.
3. Do Task Sheet 3.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 3.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

INFORMATION SHEET 3.3-1

Sowing

The process of scattering seeds in the ground soil for **growing** the crop plants. Before **sowing** good quality, seeds are selected.

Hand **sowing** or (planting) is the process of casting handfuls of seed over prepared ground, or broadcasting (from which the technological term is derived). Usually, a drag or harrow is employed to incorporate the seed into the soil. Practice is required to sow evenly and at the desired rate.

Methods of Sowing

- Traditional **Method**. A funnel-shaped tool is used to **sow** the seeds traditionally
- Broadcasting. In this process, the seeds are scattered on the seed beds either mechanically or manually
- Dibbling
- Drilling
- Seed Dropping behind the Plough
- Transplanting
- Hill Dropping
- Check Row Planting

Seed or **planting materials** of a **crop** or a **plant** is a very **important** component of **crop production** as good seed interacts with the good environment around it and determines the health of the **crop** or **plant**. If the environment is conducive and the seed is poor, **production** is likely to be less.

It typically involves (1) **plowing** to "till" or dig-up, mix, and overturn the soil; (2) **harrowing** to break the soil clods into smaller mass and incorporate **plant** residue, and (3) **leveling** the field. Initial **land preparation** begins after your last harvest or during fallow period.

Plants have simple needs: light, air, good soil and water. To grow a **plant** successfully, you must balance these resources carefully. The best way to do that is to create a controlled environment that you can adjust to keep your **plant** happy and help it grow tall.

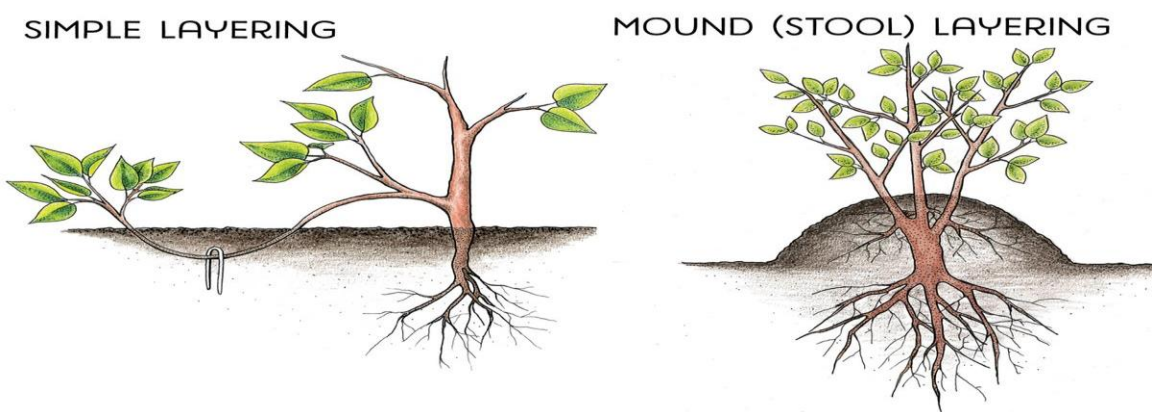
Planting material refers to the type of **material** used to establish a field or replace a **plant**. The type of **planting material** generally falls into one of two groups: conventional **planting material** and tissue-culture plantlets.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

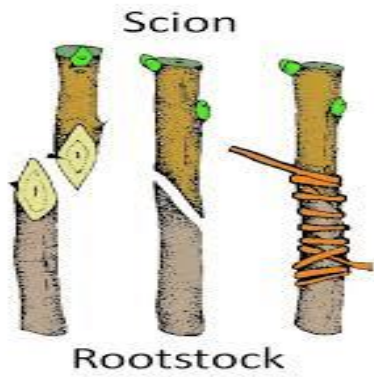
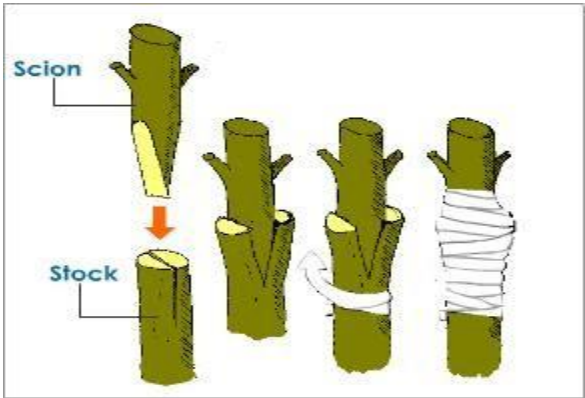


The major **methods** of asexual **propagation** are **cuttings**, layering, division, budding and grafting. **Cuttings** involve rooting a severed piece of the parent plant; layering involves rooting a part of the parent and then severing it; and budding and grafting is joining two plant parts from different varieties.

The development of roots on a stem while the stem is still attached to the parent plant is called **layering**. A **layer** is the rooted stem following detachment (removal) from the parent plant. Some **plants propagate** naturally by **layering**, but sometimes plant propagators assist the process. January 31, 1999.



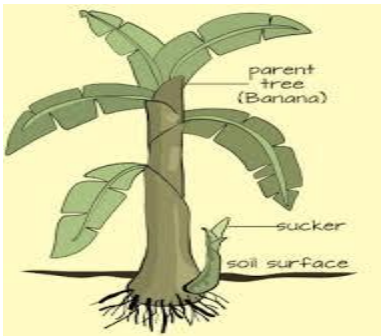
	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:		Date Developed:	
		November 16, 2020		November 16, 2020	
		Developed by:		Issued by:	Page No.
		Jumelyn P. Buya		Revision:	



Runners- horizontal running over the soil surface



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	



Root is the non-green part of the plant that normally grows under the soil and supports the plant. **Rhizome** is an underground stem meant for the storage of food. Example potato, onion, ginger.

Rhizome, also **called** creeping rootstalk, horizontal underground plant stem capable of producing the shoot and root systems of a new plant. **Rhizomes** are used to store starches and proteins and enable plants to perennate (survive an annual unfavourable season) underground.



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

SELF-CHECK 3.1-1

- 1. What are the common planting materials in crop production?
- 2. refers to the type of **material** used to establish a field or replace a **plant**.
- 3. is the rooted stem following detachment (removal) from the parent plant.
- 4. the non-green part of the plant that normally grows under the soil and supports the plant.
- 5. is an underground stem meant for the storage of food.



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

KEY ANSWER SHEET 3.1.2

- 1. The common planting materials in crop production are stem cuttings, layer, rhizome, roots, and runners
- 2. Planting materials
- 3. Layering
- 4. Root
- 5. Rhizome



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

Learning Outcome 4

LO4. Title: Prepare growing media

Learning Activities	Special Instructions
1. Read Information Sheet 4.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 4.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 2.1-1 and answer Self- Check 2.1-1 again.
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	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	



Sector: Agriculture and Fishery

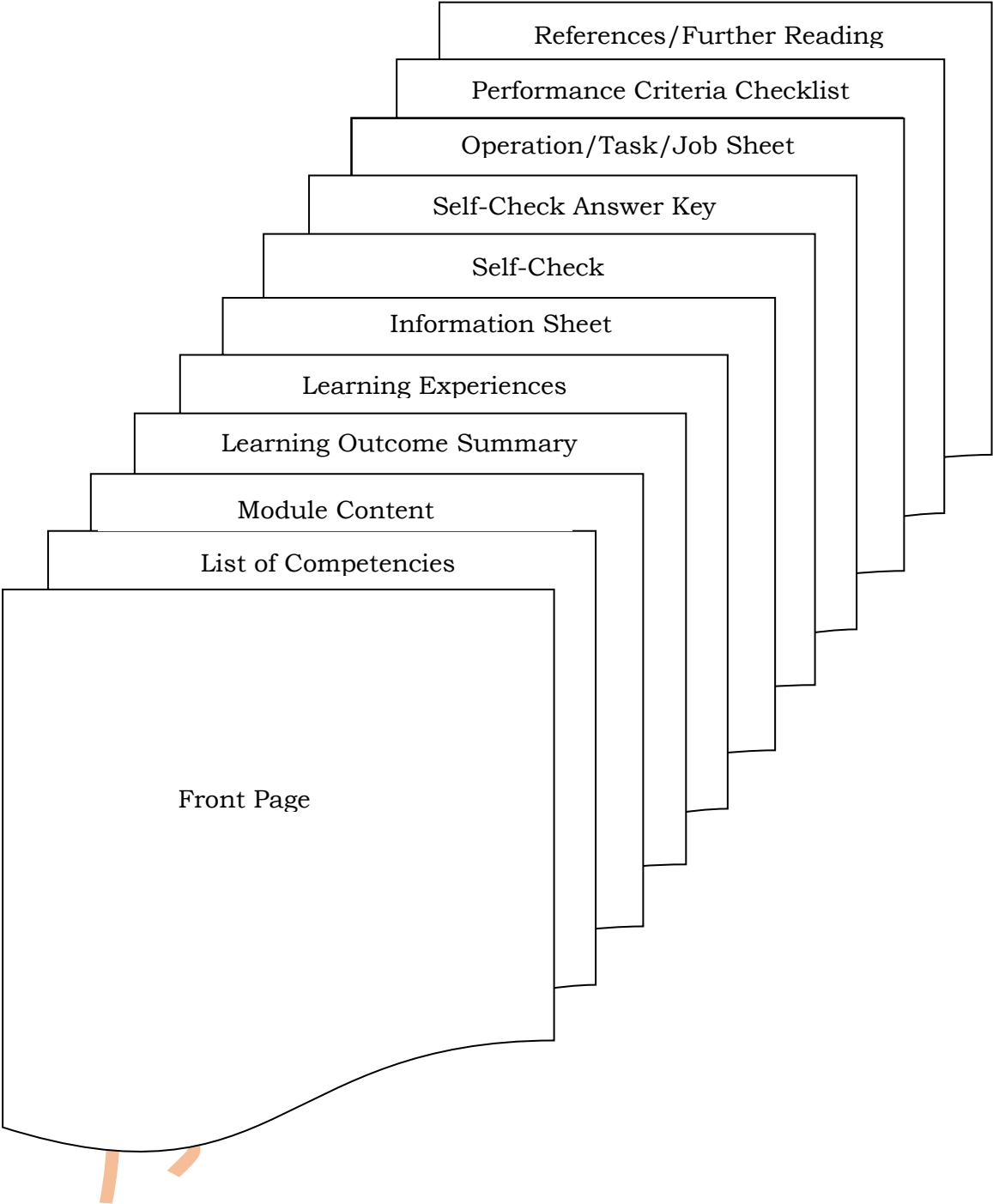
Qualification Title: Agricultural Crop production NC II

Unit of Competency: Carry Out Harvest and Post-Harvest Operation

Module Title: Carrying Out Harvest and Post-Harvest Operation

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

PARTS OF A COMPETENCY-BASED LEARNING MATERIAL



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

HOW TO USE THIS COMPETENCY-BASED LEARNING MATERIAL

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- When you feel confident that you have had sufficient practice, ask your Trainer to evaluate you. The results of your assessment will be recorded in your Progress Chart and Accomplishment Chart.

You need to complete this module before you can perform the next module.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

AGRICULTURAL CROP PRODUCTION NC II

LIST OF COMPETENCIES

No.	Unit of Competency	Module Title	Code
1.	Perform nursery operation	Performing nursery operation	AFF 610301
2.	Plant Crops	Planting crops	AFF 610302
3.	Care and maintain crops	Caring and maintaining crops	AFF 610303
4.	Carry-out harvest and post-harvest operations	Carrying-out harvest and post-harvest operations	AFF 610304

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Learning Experiences

Learning Outcome: Perform pre-harvest operations

LO1. Title: Carrying-out harvest and postharvest operations

Learning Activities	Special Instructions
1. Read Information Sheet 4.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 4.1-1.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 4.1-1 and answer Self- Check 4.1-1 again.
3. Do Task Sheet 4.1-1	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 4.1-1	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

INFORMATION SHEET 4.1.1

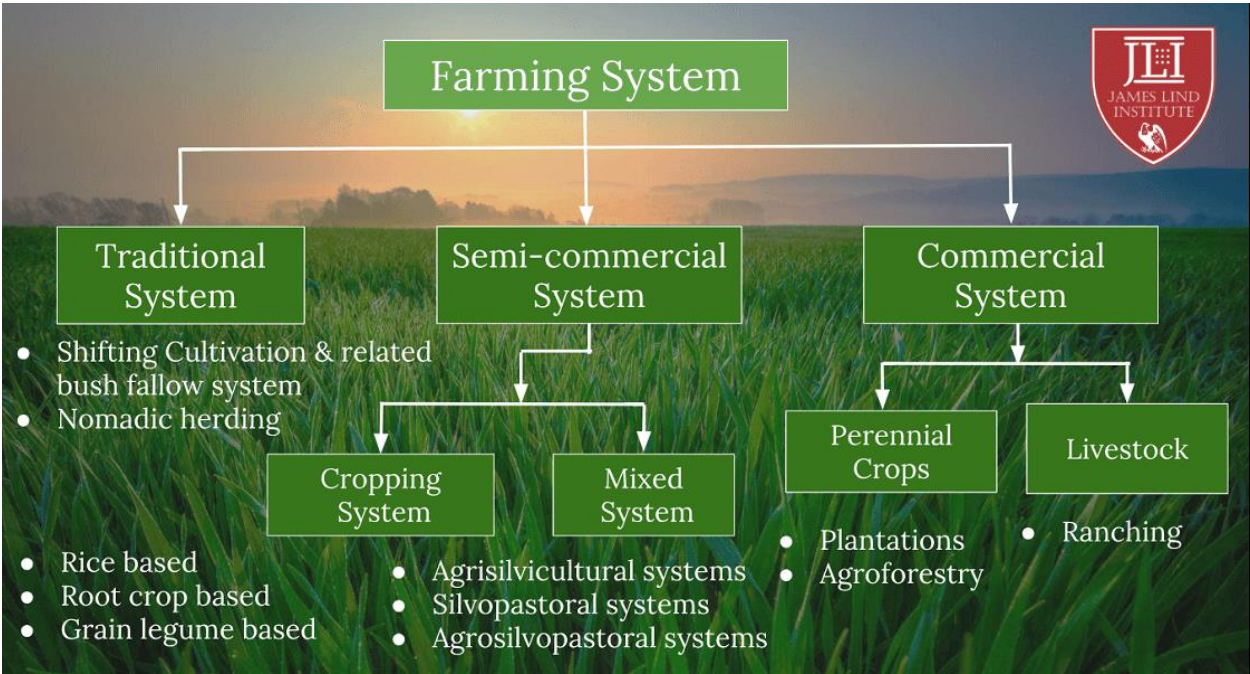
CARRY-OUT HARVEST AND POSTHARVEST OPERATIONS

Crops- A crop is a plant or animal product that can be grown and harvested extensively for profit or subsistence. Crops may refer either to the harvested parts or to the harvest in a more refined state. Most crops are cultivated in agriculture or aquaculture. A crop may include macroscopic fungus, or alga. [Wikipedia](#)

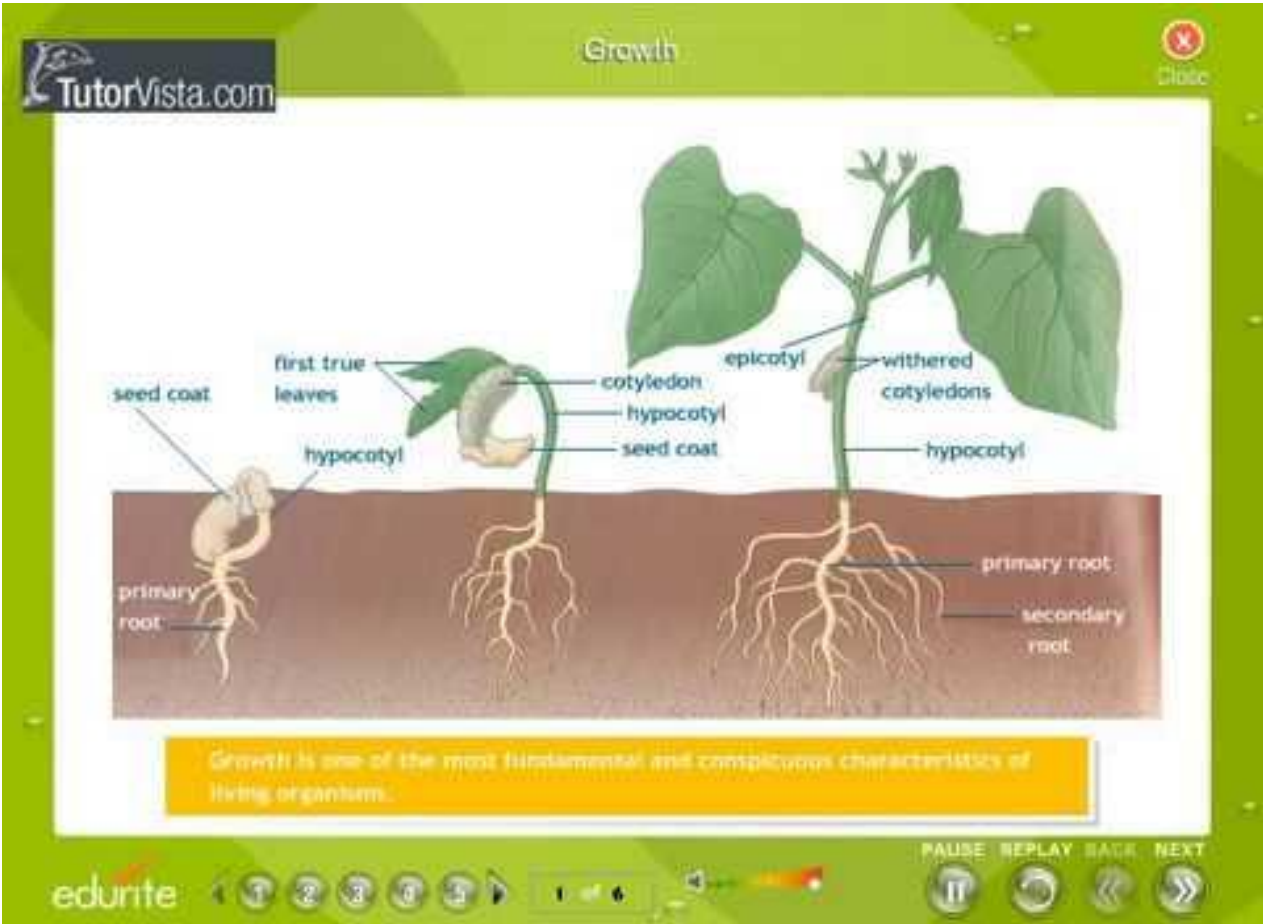
A **crop** is a plant or plant product that can be grown and harvested for profit or subsistence. By use, **crops** fall into six categories: food **crops**, feed **crops**, fiber **crops**, oil **crops**, ornamental **crops**, and industrial **crops**. Food **crops**, such as fruit and vegetables, are harvested for human consumption



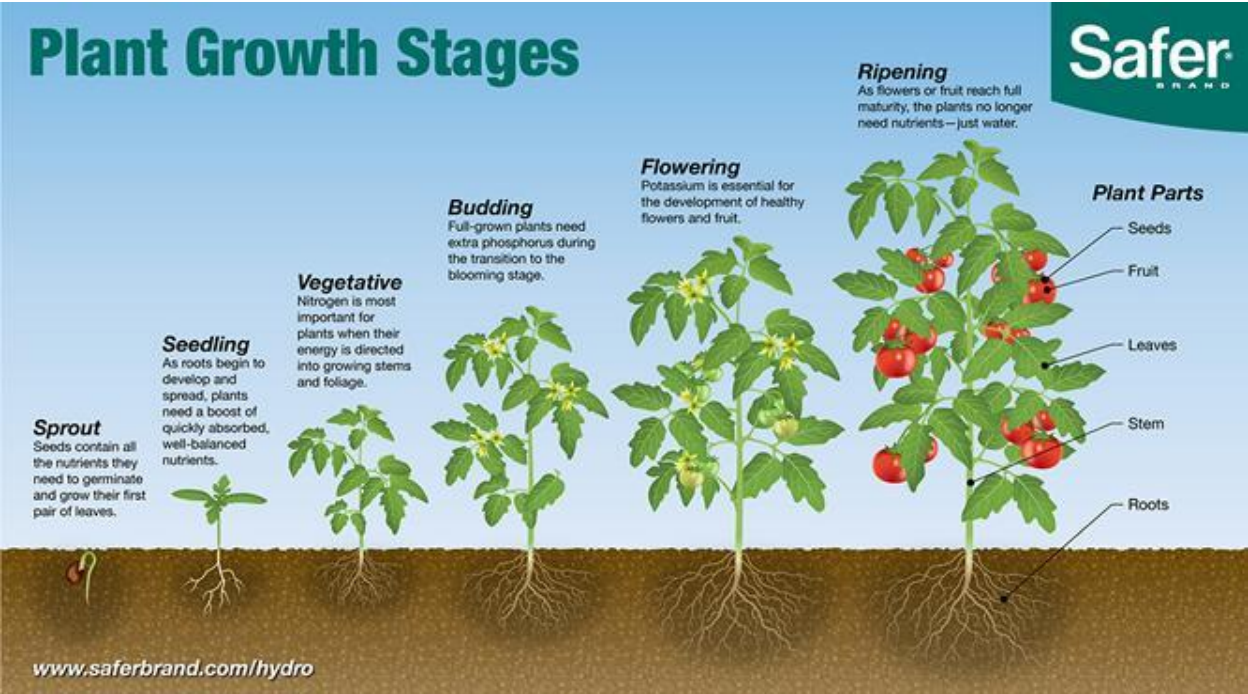
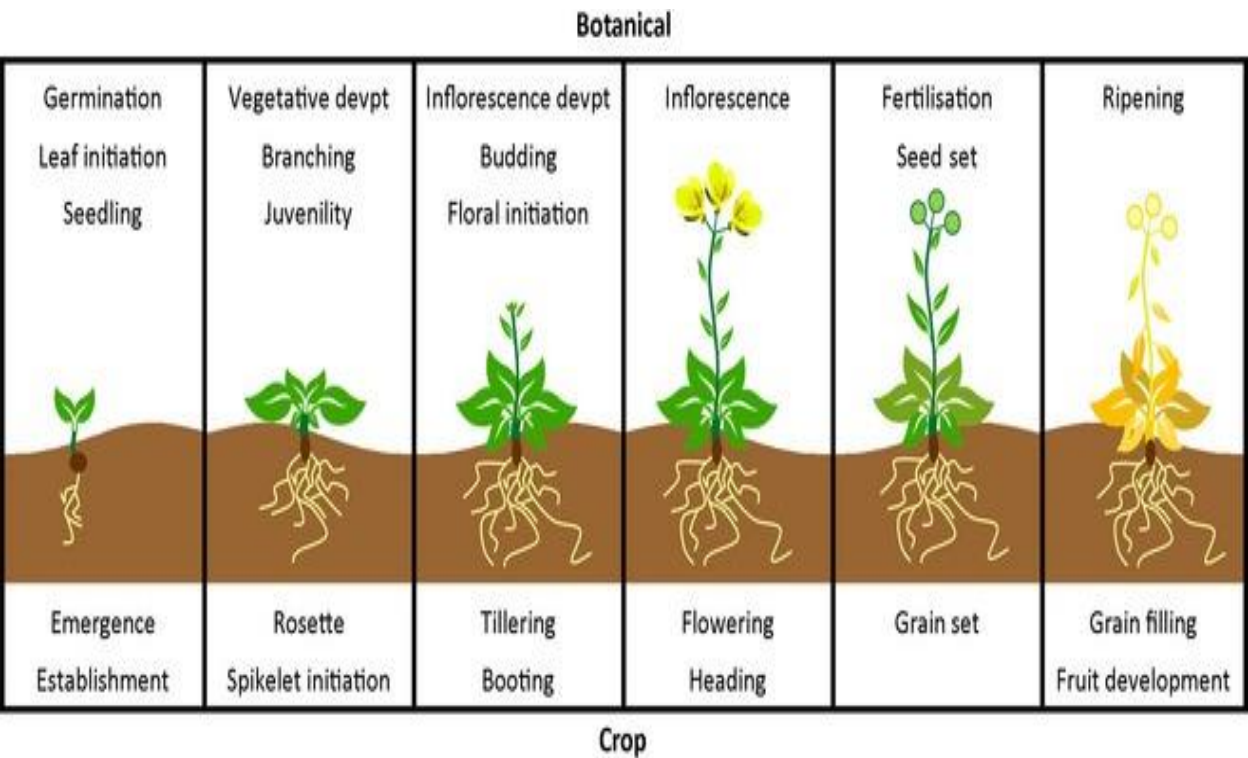
	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	



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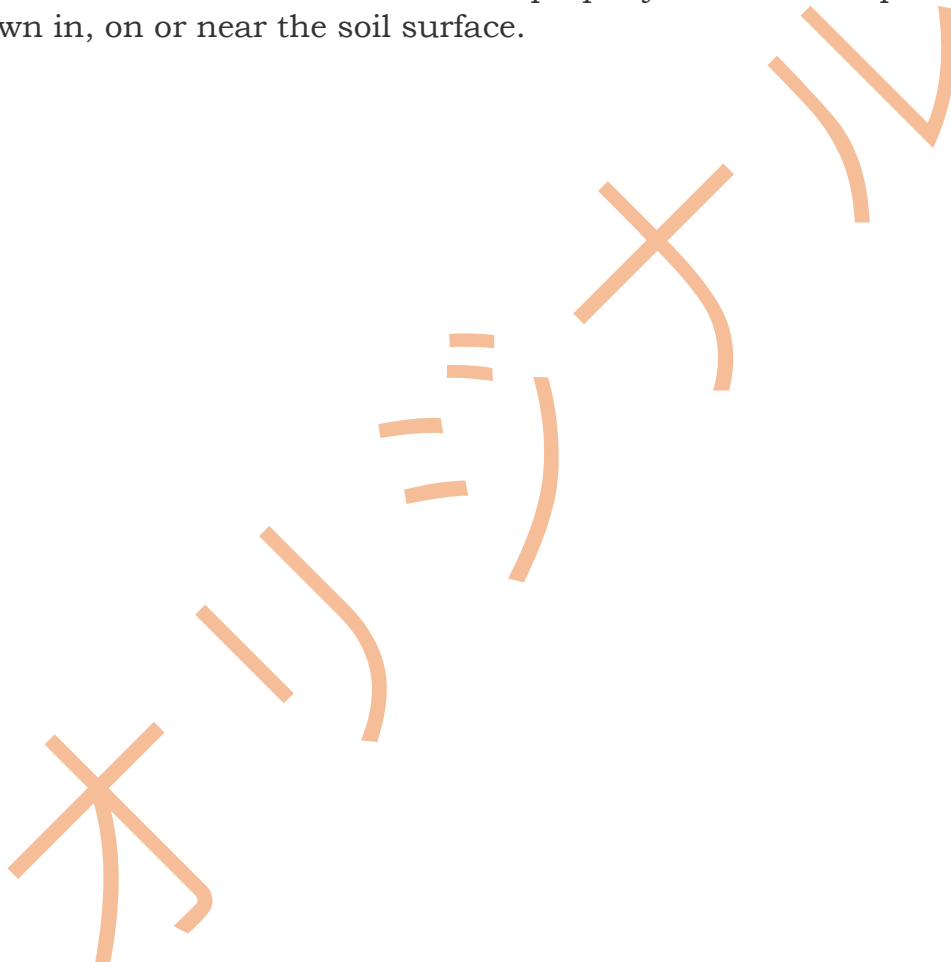
	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:
			Page No.
			Revision:



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buaya	Date Developed: November 16, 2020 Issued by: Revision:
			Page No.

Food Safety

Environmental activists and the media have raised serious concerns over the safety of our food supply. Their concerns have been focused on pesticide use on and residues in our fruits and vegetables. However, a more eminent danger lies in the potential for borne illness due to microbial contamination of fresh produce. These organisms can readily contaminate fresh produce during any number of operations associated with the production, handling, packaging, processing, distribution, and preparation of produce for consumption. As a result, significantly higher incidences of illnesses and deaths have been found to be caused by microbial contamination than from pesticide poisoning. The danger of contamination of produce by microbial human pathogens is increased with the use of untreated manures or improperly handled compost on crops grown in, on or near the soil surface.



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

Learning Outcome: Perform harvesting activity

LO2. Title: Carrying-out harvest and postharvest operations

Learning Activities	Special Instructions
1. Read Information Sheet 4.1-2 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 4.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 4.1-2 and answer Self- Check 4.1-2 again.
3. Do Task Sheet 4.1-2	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 4.1-2	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

INFORMATION SHEET 4.2.1

Harvesting processes

- Reaping - **cutting** the mature panicles and straw above ground.
- Threshing - separating the paddy grain from the rest of cut crop.
- Cleaning - removing immature, unfilled, non-grain materials.
- Hauling - moving the cut crop to the threshing location.

Hoosier Gardner: Here's how to properly harvest fruits and vegetables

How to Harvest the **vegetables**. ...

- Twist tomatoes from the vine until the **fruit** breaks free. ...
- Snip off peppers close to the stem. ...
- Snip off cucumber, squash, pumpkin and eggplant.
- Pick beans and peas by hand by pinching stems with your thumb and forefinger.

Maturity indices

Size, shape, and surface characteristics. Changes in the size, shape, or surface characteristics of **fruits and vegetables** are commonly used as **maturity indices**. For example, **vegetables** in particular are harvested when they have reached a marketable size and before they become too large.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Learning Outcome: Perform postharvest operation
 LO3. Title: Carrying-out harvest and postharvest operations

Learning Activities	Special Instructions
1. Read Information Sheet 4.1-3 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 4.1-3.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 4.1-3 and answer Self- Check 4.1-3 again.
3. Do Task Sheet 4.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 4.1-3	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:		Date Developed:	
		November 16, 2020		November 16, 2020	
		Developed by:		Issued by:	Page No.
		Jumelyn P. Buya		Revision:	

Harvesting is the process of gathering a ripe crop from the fields. Reaping is the cutting of grain or pulse for harvest, typically using a scythe, sickle, or reaper. On smaller farms with minimal mechanization, harvesting is the most labor-intensive activity of the growing season. [Wikipedia](#)



Commonly used tools **for fruit and vegetable harvesting** are secateurs or knives, and hand held or pole mounted picking shears. When **fruits** or **vegetables** are difficult to catch, such as mangoes or avocados, a cushioning material is placed around the tree to prevent damage to the **fruit** when dropping from high trees.

Techniques to extend postharvest shelf life:

- reducing respiration by lowering temperature
- slow respiration by maintaining optimal gaseous environment
- slow water loss by maintaining optimal relative humidity
- variety selection

Postharvest handling includes all steps involved in moving a commodity from the producer to the consumer including harvesting, handling, cooling, curing, ripening, packing, packaging, storing, shipping, wholesaling, retailing, and any other procedure that the product is subjected to. Because vegetables can change hands so many times in the Postharvest sector, a high level of management is necessary to ensure that quality is maintained. Each time someone fails to be conscientious in carrying out his or her assigned responsibility, quality is irreversibly sacrificed.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Table 1. Cooling Methods for Vegetables

Method ¹	Vegetable		Comments
Room Cooling	All vegetables		Too slow for many perishable commodities. Cooling rates vary extensively within loads, pallets, and containers
Forced Cooling (Pressure Cooling)	Air	Strawberry, Fruit-type Vegetables, Tubers, Cauliflower	Much faster than room cooling, cooling rates very uniform. Container venting and stacking requirements are critical to effective cooling
Hydrocooling		Stems, Leafy Vegetables, Some Fruit-type Vegetables	Very fast cooling; uniform cooling in bulk if properly used, but may vary extensively in packed shipping containers; daily cleaning and sanitation measure essential; product must tolerate wetting; water tolerant shipping containers may be needed
Package Icing		Roots, Stems, Some Flower-type Vegetables, Green Onion, Brussels Sprouts	Fast cooling; limited to commodities that can tolerate water-ice contact; water tolerant shipping containers are essential
Transit Cooling Mechanical Refrigeration		All Vegetables	
Top-Icing and Channel-Icing		Some Roots, Stems, Leafy Vegetables, Muskmelon	Cooling in most available equipment is too slow and variable; generally, not effective. Slow and irregular, top-ice weight reduces net pay load; water-tolerant shipping containers needed

Adapted from Knott’s Handbook for Vegetable Growers. 3rd Ed.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

Learning Outcome: Monitor storage pest and diseases
 LO4. Title: Carrying-out harvest and postharvest operations

Learning Activities	Special Instructions
1. Read Information Sheet 4.1-4 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 4.1-4.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 4.1-4 and answer Self- Check 4.1-4 again.
3. Do Task Sheet 4.1-4	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 4.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:		Date Developed:	
		November 16, 2020		November 16, 2020	
		Developed by:		Issued by:	Page No.
		Jumelyn P. Buya		Revision:	

INFORMATION SHEET 4.1.1

Harvesting

-process of removal of entire plants or economic parts after maturity. the economic product may be grain, seed leaf, root or entire plant. The remaining portion of the stem that is left on the field after harvest-stubble (the cut stalks of grain plants left sticking-out of the ground

-the stage of crop production immediately following harvest, including cooling, cleaning and sorting and packing. Postharvest treatment largely determines final quality, whether a crop is solid for fresh consumption, or used as an ingredient in a processed food product.

are the sign or indication the readiness of the commodity for harvest. It is the basis for determining harvest date.

Maturity-the maturity of harvested perishable commodities has an important bearing on their storage life and quality and may affect the way they are handled, transported and marketed.

The **maturity index** for a commodity is a measurement or measurements that can be used to determine whether a particular example of the commodity is mature. These **indices** are important to the trade in fresh fruits and vegetables for several reasons.

Maturity index for fruits and vegetables

Stage of maturity	Days	Ethephon doses (ppm)
Green	1	19.31 ^b
Slightly ripen	3	46.27 ^b
Slightly ripen	5	57.00 ^c
Slightly ripen	7	68.35 ^a

Maturity index for fruits and vegetables

- Skin colour: This factor is commonly applied to fruits, since skin colour changes as fruit ripens or matures
- Optical methods
- Shape
- Size
- Aroma
- Fruit opening
- Leaf changes
- Abscission

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Maturity Indices

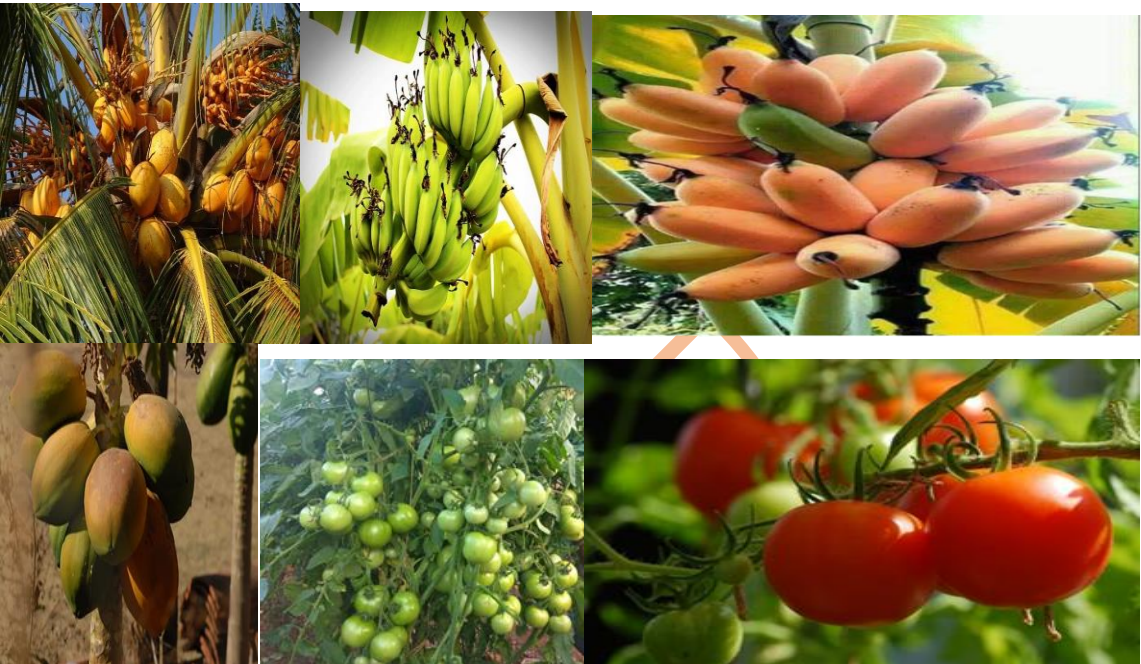
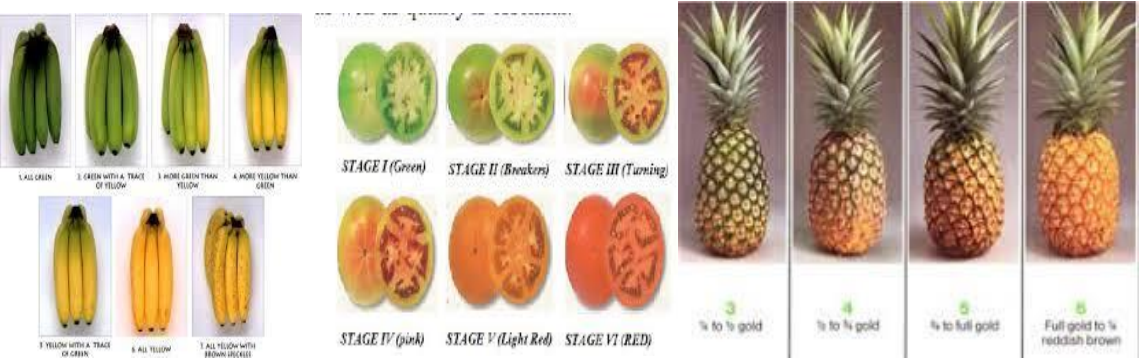


Figure 2. Coconut, banana, papaya, tomatoes, strawberries



Figure 4. Chillies, bell pepper, capsicum, cauliflower, broccoli

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	



Figure 5. Ready to harvest broccoli, Nagano, Japan



Figure 6.. Ready to harvest Chinese cabbage, Nagano Japan
 Figure 7. On-going harvesting using packaging crates directly cleaned, trimmed and sorted to high-end market (Dizon farm/ Green Star) From Jbuya Farm, La Trinidad, Benguet



Figure 8. On-going harvesting of head lettuce properly trimmed, cleaned, sorted and packed ready for shipment to JA cooperative as consolidator to Supermarkets (harvesting and post-harvest handling in Nagano, Japan)

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:		Date Developed:	
		November 16, 2020		November 16, 2020	
		Developed by: Jumelyn P. Buya		Issued by:	Page No.
				Revision:	



Figure 9. from traditional way of packaging area at Strawberry Farm to properly packed strawberry



Harvest handling
4.1.1 Maturity index for fruits and vegetables

The principles dictating at which stage of maturity a fruit or vegetable should be harvested are crucial to its subsequent storage and marketable life and quality. Post-harvest physiologists distinguish three stages in the life span of fruits and vegetables: maturation, ripening, and senescence. Maturation is indicative of the fruit being ready for harvest. At this point, the edible part of the fruit or vegetable is fully developed in size, although it may not be ready for immediate consumption. Ripening follows or overlaps maturation, rendering the produce edible, as indicated by taste. Senescence is the last stage, characterized by natural degradation of the fruit or vegetable, as in loss of texture, flavour, etc. (senescence ends at the death of the tissue of the fruit). Some typical maturity indexes are described in following sections.

Skin colour:

This factor is commonly applied to fruits, since skin colour changes as fruit ripens or matures. Some fruits exhibit no perceptible colour change during maturation, depending on the type of fruit or vegetable. Assessment of harvest maturity by skin colour depends on the judgment of the harvester, but colour charts are available for cultivars, such as apples, tomatoes, peaches, chilli peppers, etc.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Page No.
		Revision:	

Optical methods:

Light transmission properties can be used to measure the degree of maturity of fruits. These methods are based on the chlorophyll content of the fruit, which is reduced during maturation. The fruit is exposed to a bright light, which is then switched off so that the fruit is in total darkness. Next, a sensor measures the amount of light emitted from the fruit, which is proportional to its chlorophyll content and thus its maturity.

Shape:

The shape of fruit can change during maturation and can be used as a characteristic to determine harvest maturity. For instance, a banana becomes more rounded in cross-sections and less angular as it develops on the plant. Mangoes also change shape during maturation. As the mango matures on the tree the relationship between the shoulders of the fruit and the point at which the stalk is attached may change. The shoulders of immature mangoes slope away from the fruit stalk; however, on more mature mangoes the shoulders become level with the point of attachment, and with even more maturity the shoulders may be raised above this point.

Size:

Changes in the size of a crop while growing are frequently used to determine the time of harvest. For example, partially mature cobs of *Zea mays saccharata* are marketed as sweet corn, while even less mature and thus smaller cobs are marketed as baby corn. For bananas, the width of individual fingers can be used to determine harvest maturity. Usually a finger is placed midway along the bunch and its maximum width is measured with callipers; this is referred to as the calliper grade.

Aroma:

Most fruits synthesize volatile chemicals as they ripen. Such chemicals give fruit its characteristic odour and can be used to determine whether it is ripe or not. These odours may only be detectable by humans when a fruit is completely ripe, and therefore has limited use in commercial situations.

Fruit opening:

Some fruits may develop toxic compounds during ripening, such as ackee tree fruit, which contains toxic levels of hypoglycine. The fruit splits when it is fully mature, revealing black seeds on yellow arils. At this stage, it has been shown to contain minimal amounts of hypoglycine or none at all. This creates a problem in marketing; because the fruit is so mature, it will have a very short post-harvest life. Analysis of hypoglycine ‘A’ (hyp.) in ackee tree fruit revealed that the seed contained appreciable hyp. at all stages of maturity, at approximately 1000 ppm, while levels in the membrane mirrored those in the arils. This analysis supports earlier observations that unopened or partially opened ackee fruit should not be consumed, whereas fruit that opens naturally to over 15 mm of lobe separation poses little health hazard, provided the seed and membrane

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

portions are removed. These observations agree with those of Brown et al. (1992) who stated that bright red, full sized ackee should never be forced open for human consumption.

Leaf changes:

Leaf quality often determines when fruits and vegetables should be harvested. In root crops, the condition of the leaves can likewise indicate the condition of the crop below ground. For example, if potatoes are to be stored, then the optimum harvest time is soon after the leaves and stems have died. If harvested earlier, the skins will be less resistant to harvesting and handling damage and more prone to storage diseases.

Abscission:

As part of the natural development of a fruit an abscission layer is formed in the pedicel. For example, in cantaloupe melons, harvesting before the abscission layer is fully developed results in inferior flavoured fruit, compared to those left on the vine for the full period.

Firmness:

A fruit may change in texture during maturation, especially during ripening when it may become rapidly softer. Excessive loss of moisture may also affect the texture of crops. These textural changes are detected by touch, and the harvester may simply be able to gently squeeze the fruit and judge whether the crop can be harvested. Today sophisticated devices have been developed to measure texture in fruits and vegetables, for example, texture analyzers and pressure testers; they are currently available for fruits and vegetables in various forms. A force is applied to the surface of the fruit, allowing the probe of the penetrometer or *texturometer* to penetrate the fruit flesh, which then gives a reading on firmness. Hand held pressure testers could give variable results because the basis on which they are used to measure firmness is affected by the angle at which the force is applied. Two commonly used pressure testers to measure the firmness of fruits and vegetables are the Magness-Taylor and UC Fruit Firmness testers (Figure 2.1). A more elaborate test, but not necessarily more effective, uses instruments like the Instron Universal Testing Machine. It is necessary to specify the instrument and all settings used when reporting test pressure values or attempting to set standards. (texturometer- measurement of the mechanical textural parameters, hardness, cohesiveness, viscosity, elasticity, adhesiveness, brittleness, chewiness and gumminess)

Packaging

According to Wills et al. (1989), modern packaging must comply with the following requirements:

- a) The package must have sufficient mechanical strength to protect the contents during handling, transport, and stacking.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

- b) The packaging material must be free of chemical substances that could transfer to the produce and become toxic to man.
- c) The package must meet handling and marketing requirements in terms of weight, size, and shape.
- d) The package should allow rapid cooling of the contents. Furthermore, the permeability of plastic films to respiratory gases could also be important.
- e) Mechanical strength of the package should be largely unaffected by moisture content (when wet) or high humidity conditions.
- f) The security of the package or ease of opening and closing might be important in some marketing situations.
- g) The package must either exclude light or be transparent.
- h) The package should be appropriate for retail presentations.
- i) The package should be designed for ease of disposal, re-use, or recycling.
- j) Cost of the package in relation to value and the extent of contents protection required should be as low as possible.

Classification of packaging:

Packages can be classified as follows:

- Flexible sacks; made of plastic jute, such as bags (small sacks) and nets (made of open mesh)
- Wooden crates
- Cartons (fibreboard boxes)
- Plastic crates
- Pallet boxes and shipping containers
- Baskets made of woven strips of leaves, bamboo, plastic, etc.

Uses for above packages:

Nets are only suitable for hard produce such as coconuts and root crops (potatoes, onions, yams).

Wooden crates are typically wire bound crates used for citrus fruits and potatoes, or wooden field crates used for softer produce like tomatoes. Wooden crates are resistant to weather and more efficient for large fruits, such as watermelons and other melons, and generally have good ventilation. Disadvantages are that rough surfaces and splinters can cause damage to the produce, they can retain undesirable odours when painted, and raw wood can easily become contaminated with moulds.

Fiberboard boxes are used for tomato, cucumber, and ginger transport. They are easy to handle, light weight, come in different sizes, and come in a variety of colours that can make produce more attractive to consumers. They have some disadvantages, such as the effect of high humidity, which

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

can weaken the box; neither are they waterproof, so wet products would need to be dried before packaging. These boxes are often of lower strength compared to wooden or plastic crates, although multiple thickness trays are very widely used. They can come flat packed with ventilation holes and grab handles, making a cheap attractive alternative that is very popular. Care should be taken that holes on the surface (top and sides) of the box allow adequate ventilation for the produce and prevent heat generation, which can cause rapid product deterioration.

Plastic crates are expensive but last longer than wooden or carton crates.

They are easy to clean due to their smooth surface and are hard in strength, giving protection to products. Plastic crates (Figure 2.8) can be used many times, reducing the cost of transport. They are available in different sizes and colours and are resistant to adverse weather conditions. However, plastic crates can damage some soft produce due to their hard surfaces, thus liners are recommended when using such crates.

Pallet boxes are very efficient for transporting produce from the field to the packinghouse or for handling produce in the packinghouse. Pallet boxes have a standard floor size (1200 × 1000 mm) and depending on the commodity have standard heights. Advantages of the pallet box are that it reduces the labour and cost of loading, filling, and unloading; reduces space for storage; and increases speed of mechanical harvest. The major disadvantage is that the return volume of most pallet boxes is the same as the full load. Higher investment is also required for the forklift truck, trailer, and handling systems to empty the boxes. They are not affordable to small producers because of high, initial capital investment.

Typical plastic crate holding fresh oranges.



4.1.4 Cooling methods and temperatures

Several methods of cooling are applied to produce after harvesting to extend shelf life and maintain a fresh-like quality. Some of the low temperature treatments are unsuitable for simple rural or village treatment but are included for consideration as follows:

Precooling

Fruit is precooled when its temperature is reduced from 3 to 6°C (5 to 10°F) and is cool enough for safe transport. Precooling may be done with cold

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

air, cold water (hydrocooling), direct contact with ice, or by evaporation of water from the product under a partial vacuum (vacuum cooling). A combination of cooled air and water in the form of a mist called hydrocooling is an innovation in cooling of vegetables.

Air precooling

Precooling of fruits with cold air is the most common practice. It can be done in refrigerator cars, storage rooms, tunnels, or forced air-coolers (air is forced to pass through the container via baffles and pressure differences).

Icing

Ice is commonly added to boxes of produce by placing a layer of crushed ice directly on the top of the crop. An ice slurry can be applied in the following proportion: 60% finely crushed ice, 40% water, and 0.1% sodium chloride to lower the melting point. The water to ice ratio may vary from 1:1 to 1:4.

Room cooling

This method involves placing the crop in cold storage. The type of room used may vary, but generally consists of a refrigeration unit in which cold air is passed through a fan. The circulation may be such that air is blown across the top of the room and falls through the crop by convection. The main advantage is cost because no specific facility is required.

Forced air-cooling

The principle behind this type of precooling is to place the crop into a room where cold air is directed through the crop after flowing over various refrigerated metal coils or pipes. Forced air-cooling systems blow air at a high velocity leading to desiccation of the crop. To minimize this effect, various methods of humidifying the cooling air have been designed such as blowing the air through cold water sprays.

Hydrocooling

The transmission of heat from a solid to a liquid is faster than the transmission of heat from a solid to a gas. Therefore, cooling of crops with cooled water can occur quickly and results in zero loss of weight. To achieve high performance, the crop is submerged in cold water, which is constantly circulated through a heat exchanger. When crops are transported around the packhouse in water, the transport can incorporate a hydrocooler. This system has the advantage wherein the speed of the conveyer can be adjusted to the time required to cool the produce. Hydrocooling has a further advantage over other precooling methods in that it can help clean the produce. Chlorinated water can be used to avoid spoilage of the crop. Hydrocooling is commonly used for vegetables, such as asparagus, celery, sweet corn, radishes, and carrots, but it is seldom used for fruits.

Vacuum cooling

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Cooling in this case is achieved with the latent heat of vaporization rather than conduction. At normal air pressure (760 mmHg) water will boil at 100°C. As air pressure is reduced so is the boiling point of water, and at 4.6 mmHg water boils at 0°C. For every 5 or 6°C reduction in temperature, under these conditions, the crop loses about 1% of its weight (Barger, 1961). This weight loss may be minimized by spraying the produce with water either before enclosing it in the vacuum chamber or towards the end of the vacuum cooling operation (hydrovacuum cooling). The speed and effectiveness of cooling is related to the ratio between the mass of the crop and its surface area. This method is particularly suitable for leaf crops such as lettuce. Crops like tomatoes having a relatively thick wax cuticle are not suitable for vacuum cooling.

Recommended minimum temperature to increase storage time

There is no ideal storage for all fruits and vegetables, because their response to reduced temperatures varies widely. The importance of factors such as mould growth and chilling injuries must be taken into account, as well as the required length of storage (Wills et al., 1989). Storage temperature for fruits and vegetables can range from -1 to 13°C, depending on their perishability. Extremely perishable fruits such as apricots, berries, cherries, figs, watermelons can be stored at -1 to 4°C for 1-5 weeks; less perishable fruits such as mandarin, nectarine, ripe or green pineapple can be stored at 5-9°C for 2-5 weeks; bananas at 10°C for 1-2 weeks and green bananas at 13°C for 1-2 weeks. Highly perishable vegetables can be stored up to 4 weeks such as asparagus, beans, broccoli, and Brussels sprouts at -1-4°C for 1-4 weeks; cauliflower at 5-9°C for 2-4 weeks. Green tomato is less perishable and can be stored at 10°C for 3-6 weeks and non-perishable vegetables such as carrots, onions, potatoes and parsnips can be stored at 5-9°C for 12-28 weeks. Similarly, sweet potatoes can be stored at 10°C for 16-24 weeks. The storage life of produce is highly variable and related to the respiration rate; there is an inverse relation between respiration rate and storage life in that produce with low respiration generally keeps longer.

For example, the respiration rate of a very perishable fruit like ripe banana is 200 mL CO₂.kg⁻¹h⁻¹ at 15°C, compared to a non-perishable fruit such as apple, which has a respiration rate of 25 mL CO₂.kg⁻¹ h⁻¹ at 15°C.

2. High temperatures

Exposure of fruits and vegetables to high temperatures during post-harvest reduces their storage or marketable life. This is because as living material, their metabolic rate is normally higher with higher temperatures. High temperature treatments are beneficial in curing root crops, drying bulb crops, and controlling diseases and pests in some fruits. Many fruits are exposed to high temperatures in combination with ethylene (or another suitable gas) to initiate or improve ripening or skin colour.

Pest control and decay

Crops may be immersed in hot water before storage or marketing to control disease. A common disease of fruits known as anthracnose, caused by the

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

infection of fungus *Colletotrychum spp.* can be successfully controlled in this way. Combining appropriate doses of fungicides with hot water is often effective in controlling disease in fruits after harvesting.

HARVESTING

The goals of harvesting are to gather a commodity from the field at the proper level of maturity with a minimum of damage and loss, as rapidly as possible and at a minimum cost. This is achieved through hand-harvesting in most fruit, vegetable and flower crops. 1. Hand Harvesting has a number of advantages over machine harvest. People can accurately determine product quality, allowing accurate selection of mature product. This is particularly important for crops that have a wide range of maturity and need to be harvested several times during the season. Properly trained workers can pick and handle the product with a minimum of damage. Many fresh-market products have a short shelf life if they are bruised or damaged during harvest and handling. The rate of harvest can easily be increased by hiring more workers. Hand- harvesting also requires a minimum of capital investment. The main problem with hand harvesting is labor management. Labor supply is a problem for growers who cannot offer a long employment season. Labor strikes during the harvest period can be costly. In spite of these problems, quality is so important to marketing fresh- market commodities successfully that hand harvesting remains the dominant method of harvest of most fruits and vegetables and for all cut flowers. Effective use of hand labor requires careful management. New employees must be trained to harvest the product at the required quality and at an acceptable rate of productivity. Employees must know what level of performance and must be encouraged and trained to reach that level.

2. Mechanical harvesting is currently used for fresh-market crops that are roots, tubers, or rhizomes and for nut crops. Vegetables that are grown below ground (radishes, potatoes, garlic, carrots, beets and others) are always harvested only once and the soil can be used to cushion the product from machine caused mechanical injury. Tree nuts and peanuts are protected by a shell and easily withstand mechanical handling. A number of products destined for processing such as tomatoes, wine grapes, beans, peas, prunes, peaches and some leafy green vegetables are machine harvested because harvest damage does not significantly affect the quality of processed product. This is often because the product is processed quickly after harvest. These crops have also been amenable to new production techniques and breeding that allow the crop to be better suited to mechanical harvest. The main advantage of mechanical harvest equipment is that machines can often harvest at high rates. Tree nut harvesters, for eg. attaching a shaking mechanism to the tree and remove most of the nuts in few seconds. The nuts are either caught on a fabric-covered frame or picked up from the ground by other machines. This allows an orchard to be harvested very quickly compared to handshaking with poles. Machine harvest also reduces management problems associated with workers. The commodity must be grown to accept mechanical harvest. Demerits of Mechanical Harvesting Machines are rarely capable of selective harvest. Mechanical harvesting will not be feasible until the crop or production techniques can be modified to allow

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

one-time harvest. Harvesting machines often causes excessive product perennial crops eg. Bark damage from a tree shaker. The harvesting machines are quite expensive. III.

POST HARVEST HANDLING

Being living organs, fruits and vegetables continue to respire even after harvesting when they have a limited source of food reserves. In addition to degradation of respiratory substrates, a number of changes in taste, colour, flavour, texture and appearance take place in the harvested commodities which make them unacceptable for consumption by the consumers if these are not handled properly. **Post-harvest technology starts immediately after the harvest of fruits and vegetables.** The whole process of processing the commodities is categorized as Handling of fresh produce. Post-harvest Technology of fresh fruits and vegetables combines the biological and environmental factors in the process of value addition of a commodity.

1. Pre-cooling (prompt cooling after harvest) is important for most of the fruits and vegetables because they may deteriorate as much in 1 hour at 32°C. In addition to removal of field heat from commodities, precooling also reduces bruise damage from vibration during transit. Cooling requirement for a crop vary with the air temperature during harvesting, stage of maturity and nature of crop. There are many methods of precooling viz, cold air (room cooling, forced air cooling), cold water (hydrocooling), direct contact with ice (contact icing), evaporation of water from the produce (evaporative cooling, vacuum cooling) and combination of vacuum and hydrocooling (hydrovac cooling). Some chemicals (nutrients/growth regulators/ fungicides) can also be mixed with the water used in hydrocooling to prolong the shelf life by improving nutrient status of crop and preventing the spread of post-harvest diseases.

2. Washing, Cleaning and Trimming Before fresh fruits and vegetables are marketed various amounts of cleaning are necessary which typically involves the removal of soil dust, adhering debris, insects and spray residues. Chlorine in fresh water is often used as disinfectant to wash the commodity. Some fungicides like Diphenylamine (0.1 - 0.25%) or ethoxyquin (0.2 - 0.5%) may be used as post-harvest dip to control the disorders. Eg. Apple superficial scald. For cleaning of some fruit type vegetables (melons, brinjals, tomatoes, cucumber) they should be wiped with damp cloth. Many vegetables need trimming, cutting and removal of unsightly leaves or other vegetative parts.

3. Sorting, Grading and Sizing is done by hand to remove the fruits which are unsuitable to market or storage due to damage by insects, diseases or mechanical injuries. The remainder crop product is separated into two or more grades on the basis of the surface colour, shape or visible defects. For eg, in an apple packing house in India 3 grades viz. Extra Fancy, Fancy and standard may be packed for marketing. The fourth “cull” grade is meant for processing. After sorting and grading, sizing is done either by hand or machine. Machine sizers work on two basic principles: weight and diameter. Sizing on the basis of fruit shape and size are most effective for spherical (Oranges, tomato, certain apple cultivars) and elongated

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

(Delicious apples and European pears or of non-uniform shape) commodities, respectively.

4. Curing is an effective operation to reduce the water loss during storage from hardy vegetables viz, onion, garlic, sweet potato and other tropical root vegetables. The curing methods employed for root crops are entirely different than that from the bulbous crops (onions and garlic). The curing of root and tuber crops develops periderms over cut, broken or skinned surfaces wound restoration. It helps in the healing of harvest injuries, reduces loss of water and prevents the infection by decay pathogens. Onions and garlic are cured to dry the necks and outer scales. For the curing of onion and garlic, the bulbs are left in the field after harvesting under shade for a few days until the green tops, outer skins and roots are fully dried.

5. Waxing Quality retention is a major consideration in modern fresh fruit marketing system. Waxes are esters of higher fatty acid with monohydric alcohols and hydrocarbons and some free fatty acids. But coating applied to the surface of fruit is commonly called waxes whether or not any component is actually a wax. Waxing generally reduces the respiration and transpiration rates, but other chemicals such as fungicides, growth regulators, preservative can also be incorporated specially for reducing microbial spoilage, sprout inhibition etc. However, it should be remembered that waxing does not improve the quality of any inferior horticulture product but it can be a beneficial adjunct to good handling. The advantages of wax application are: - Improved appearances of fruit. - Reduced moisture losses and retards wilting and shriveling during storage of fruits. - Less spoilage especially due to chilling injury and browning. - Creates diffusion barrier as a result of which it reduces the availability of O₂ to the tissues thereby reducing respiration rate. - Protects fruits from micro-biological infection. - Considered a cost-effective substitute in the reduction of spoilage when refrigerated storage is unaffordable. - Wax coating are used as carriers for sprout inhibitors, growth regulators and preservatives. The principal disadvantage of wax coating is the development of off- flavour if not applied properly. Adverse flavour changes have been attributed to inhibition of O₂ and CO₂ exchange thus, resulting in anaerobic respiration and elevated ethanol and acetaldehyde contents. Paraffin wax, Carnauba wax, Bee wax, Shellac, Wood resins and Polyethylene waxes used commercially.

6. Packaging Proper or scientific packaging of fresh fruits and vegetables reduces the wastage of commodities by protecting them from mechanical damage, pilferage, dirt, moisture loss and other undesirable physiological changes and pathological deterioration during the course of storage, transportation and subsequent marketing. For providing, uniform quality to packed produce, the commodity should be carefully supervised and sorted prior to packaging. Packaging cannot improve the quality but it certainly helps in maintaining it as it protects produce against the hazards of journey. Striking developments have been in the field of packaging of horticultural produce and the gunny bags, grasses and stem leaves used so far for packaging are now being replaced by a variety of containers such as wooden boxes, baskets woven from bamboo or twigs, sack/jute bags and corrugated fibre board (CFB) boxes.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

7. Storage A number of storage techniques (ground storage, ambient storage, refrigerated storage, air cooled storage, zero energy storage, modified atmospheric storage, hypobaric storage and controlled atmosphere storage) are being used for fruits and vegetables depending upon the nature of the commodity and the storage period intended.



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

JOB SHEET 4.2-1

Title: Harvesting	
Performance Objective: Actual-hands on harvesting following the maturity indices and standard of proper handling of harvest.	
Materials needed: Harvesting knives Plastic crates Weighing balance Complete set of PPE	Procedure: 1. Prepare all materials, tools and equipment 2. Select the maturity indices indicated on the market requirement. 3. Do pre-harvesting techniques 4. Use harvesting knives and clean crates. Avoid contact to the soil. 5. Do proper sorting and grading according to market preferences. 6. Packed in the packaging crates or boxes. Always observe cleanliness by wearing appropriate PPE.
Assessment Method:	
•Written Examination •Actual-hands on •Oral Interview	

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

PERFORMANCE CHECKLIST

Criteria	YES	NO
Did I...		
1. Prepare all the materials, tools and equipment needed in harvesting?		
2. Identify the maturity indices of the vegetable and fruits?		
3. Clean and trim?		
4. Proper sorting and grading of the harvest?		
5. Pack using the plastic crates?		
6. Hauling and transporting		
7. Pre-cooling		
Name of Trainee: _____		
Date of Performance: _____		
Evaluated by: _____		

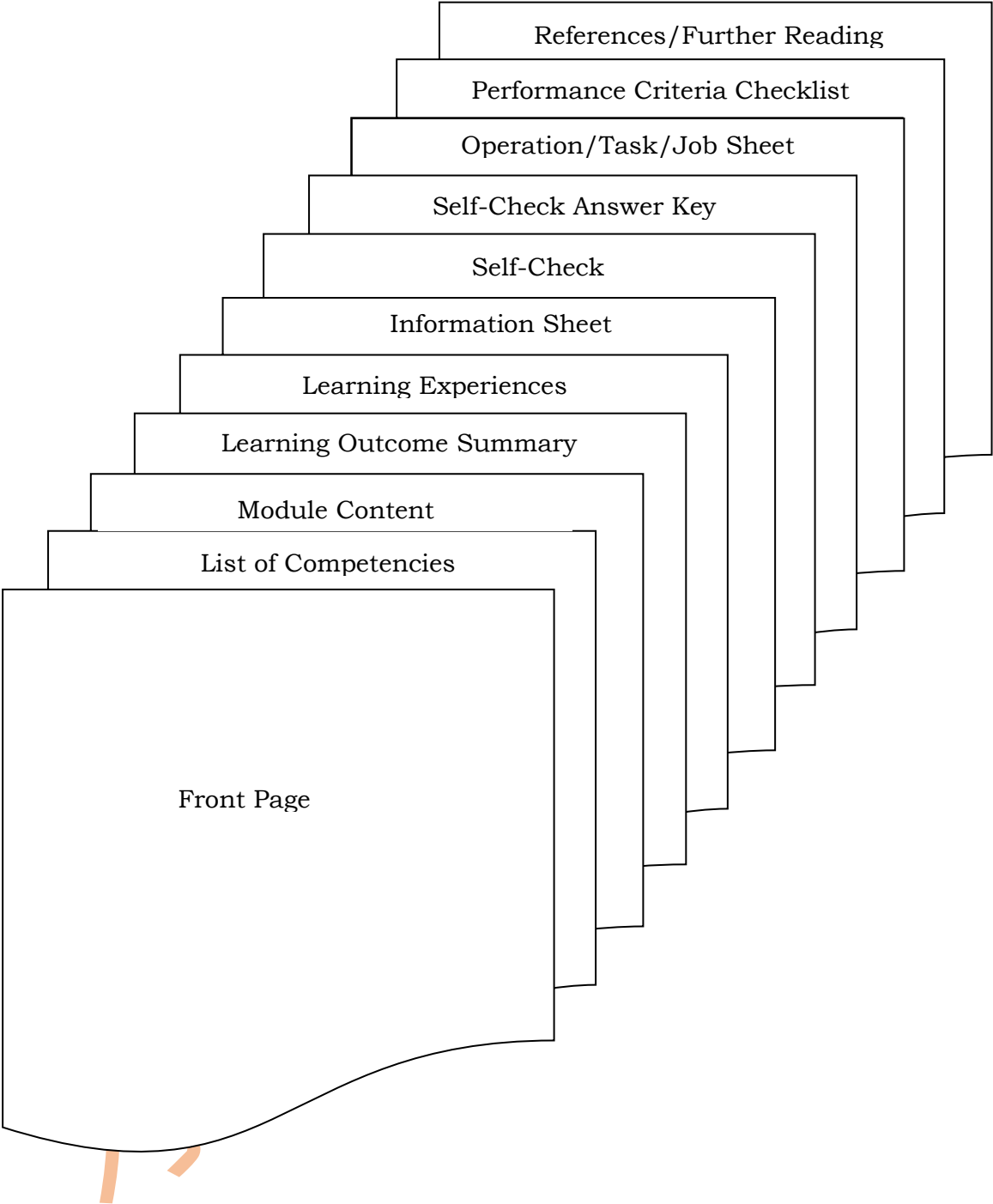
COMPETENCY BASED LEARNING MATERIAL



Sector:	Agriculture and Fishery
Qualification Title:	Agricultural Crop Production NC II
Unit of Competency:	Practice Career Professionalism
Module Title:	Practicing Career professionalism

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

PARTS OF A COMPETENCY-BASED LEARNING MATERIAL



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

HOW TO USE THIS COMPETENCY-BASED LEARNING MATERIAL

Welcome!

The unit of competency, “Practice Career Professionalism”, is one of the Common Competency of Agricultural Crop Production NC II, a course which comprises the knowledge, skills and attitudes required for National Certificate level II.

The module, Practice Career Professionalism, contains training materials and activities related to enhancing the knowledge, skills and attitude of the trainees in agricultural crop for you to complete.

In this module, you are required to go through a series of learning activities in order to complete each learning outcome. In each learning outcome are Information Sheets, Self-Checks, Operation Sheets and Task/Job Sheets. Follow and perform the activities on your own. If you have questions, do not hesitate to ask for assistance from your facilitator.

Remember to:

- Work through all the information and complete the activities in each section.
- Read information sheets and complete the self-check. Suggested references are included to supplement the materials provided in this module.
- Most probably, your trainer will also be your supervisor or manager. He is there to support you and show you the correct way to do things.
- You will be given plenty of opportunities to ask questions and practice on the job. Make sure you practice your new skills during regular work shifts. This way, you will improve your speed, memory and your confidence.
- Use the Self-Checks, Operation Sheets or Task or Job Sheets at the end of each section to test your own progress. Use the Performance Criteria Checklist or procedural Checklist located after the sheet to check your own performance.
- When you feel confident that you have had sufficient practice, ask your Trainer to evaluate you. The results of your assessment will be recorded in your Progress Chart and Accomplishment Chart.

You need to complete this module before you can perform the next module.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

AGRICULTURAL CROP PRODUCTION NC II

List of Competencies

No.	Unit of Competency	Module Title	Code
1	Participate in workplace communication	Participating in workplace communication	
2	Work in a team	Working in a team	
3	Practice career professionalism	Practicing career professionalism	
4	Practice occupational health and safety procedure	Practice occupational health and safety procedure	



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

MODULE CONTENT

Unit of Competency: Practice Career Professionalism
Module Title: Practicing Career Professionalism

Module Description:
This course is designed to enhanced the knowledge, desirable attitudes and skills of performing nursery operations, planting crops, caring and maintaining crops and carrying-out harvest and postharvest operations. It also includes competencies in workplace communication, teamwork, safety in the use of hand tools and equipment, housekeeping and processing of farm wastes.

NOMINAL DURATION: 18 hours

Unit of Competency: Practice Career Professionalism
Module Title: Practicing Career professionalism

Module Descriptor:
The unit covers the knowledge, skills and attitude in promoting career growth and advancement

Nominal Duration: 5 hours

- Summary of Learning Outcomes
- LO1. Integrate personal objectives with organizational goals
 - LO2. Set and meet work priorities
 - LO3. Maintain professional growth and development

Details of Learning Outcomes
LO1. Integrate personal objectives with organizational goals

- Assessment Criteria
- 1. Personal growth and work plans are pursued towards improving the qualifications set for the profession.
 - 2. Intra-and interpersonal relationships is maintained in the course of managing oneself based on performance evaluation.
 - 3. Commitment to the organization and its goal is demonstrated in the performance duties.

- Contents
- 1. Personal development, social aspects; intra and interpersonal development
 - 2. Organizational goals
 - 3. Personal hygiene and practices
 - 4. Code of ethics

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Conditions

- 1. Workplace
- 2. Code of ethics
- 3. Organizational goals
- 4. Hand-outs and personal development-social aspects

Methodologies

- 1. Interactive lecture
- 2. Demonstration
- 3. Simulation
- 4. Self-paced instruction

Assessment Methods

- 1. Portfolio Assessment
- 2. Interview
- 3. Written Examination
- 4. Simulation/Role plat
- 5. Observations
- 6. Third party reports

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Learning Experiences

Learning Outcome 1

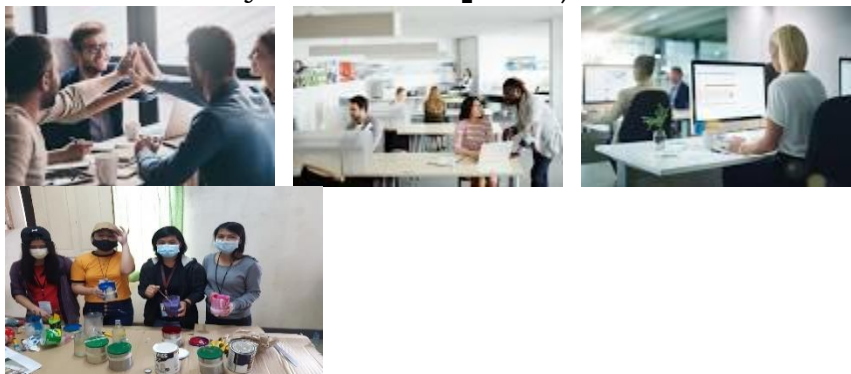
LO1. Title: Integrate personal objectives with organizational goals

Learning Activities	Special Instructions
1. Read Information Sheet 1.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 1.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 1.1-1 and answer Self- Check 1.1-1 again.
3. Do Task Sheet 1.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 1.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

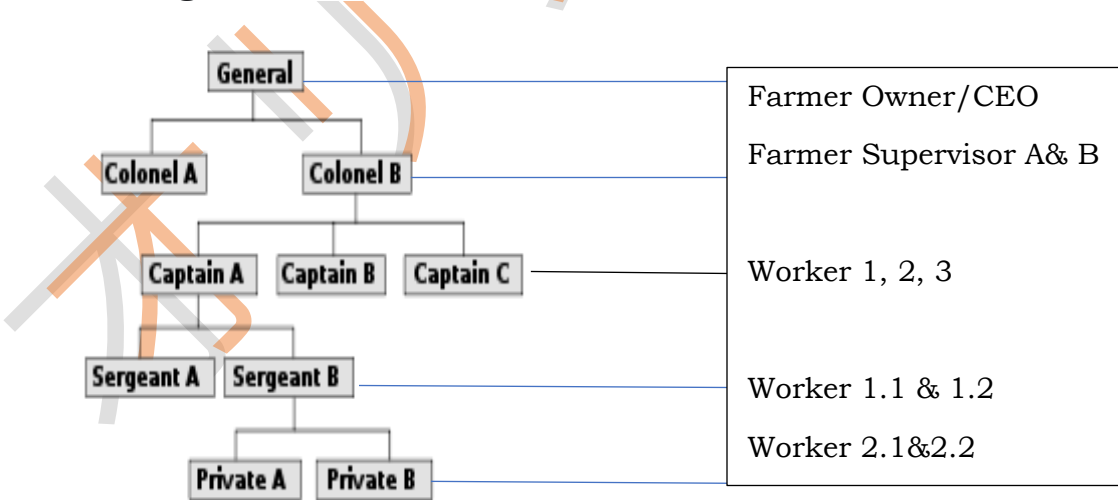
	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

INFORMATION SHEET 3.1-1

1. A place where people work, such as an office or factory ("health and safety in **the workplace**").



2. Organization- An organization, or organization, is an entity – such as a company, an institution, or an association – comprising one or more people and having a particular purpose. The word is derived from the Greek word organon, which means tool or instrument, musical instrument, and organ. [Wikipedia](#)
3. Organizational chart- An **organizational chart** is a **diagram** that visually conveys a company's internal structure by detailing the roles, responsibilities, and relationships between individuals within an entity. **Organizational charts** either broadly depict an enterprise company-wide or drill down to a specific department or unit. **Organizational charts** (or **hierarchy charts**) are the graphical representation of an **organization's structure**. Its **purpose** is to illustrate the reporting relationships and chains of command within the **organization**.



4. Personal growth- **means** different things to different people. In general, it refers to the **self-improvement** of your skills, knowledge, **personal** qualities, life goals and outlook.
5. Organizational goal- that the **organization** tries to achieve, intentions on which the **organization's** decisions and actions are based.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	



6. Code of Ethics-

code of ethics

A written set of guidelines issued by an organization to its workers and management to help them conduct their actions ...

BD BusinessDictionary

CODE OF ETHICS

- ① CLIENT SERVICE
- ② CONFLICT OF INTEREST
- ③ GROUP AUTONOMY
- ④ PRINCIPLES, METHODS AND TOOLS
- ⑤ RESPECT, SAFETY, EQUITY AND TRUST
- ⑥ STEWARDSHIP OF PROCESS
- ⑦ CONFIDENTIALITY
- ⑧ PROFESSIONAL DEVELOPMENT

THE PRINCIPLES

- 1 Responsibility to the Profession**
Trust in the educational system depends upon a level of professional conduct and responsibility that may be higher than required by law. This entails holding one another to the same ethical standards.
- 2 Responsibility for Professional Competence**
Commitment to the highest levels of professional and ethical practice, including demonstration of the knowledge, skills, and dispositions required for professional competence.
- 3 Responsibility to Students**
A primary obligation to treat students with dignity and respect, including promoting the health, safety and well-being of students by establishing and maintaining appropriate verbal, physical, emotional, and social boundaries.
- 4 Responsibility to the School Community**
Promotion of positive relationships and effective interactions with all members of the school community, while maintaining professional boundaries.
- 5 Responsible and Ethical Use of Technology**
Consideration of the impact of consuming, creating, distributing and communicating information through all technologies. Vigilance to ensure that appropriate boundaries of time, place, and role are maintained when using electronic communication.

7. Personal hygiene-

What is Personal Hygiene?

Regular Routine of Personal Care

Washing and Grooming of

- Hair
- Face and Skin
- Teeth
- Ears
- Hands
- Nails
- Feet

Personal hygiene - Definition

- Personal hygiene may be described as the principle of maintaining cleanliness and grooming of the external body.
- Failure to keep up a standard of hygiene can have many implications. Not only is there an increased risk of getting an infection or illness, but there are many social and psychological aspects that can be affected.

8. Professionalism

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

WHAT IS PROFESSIONALISM IN THE WORKPLACE?

professionalism

has to do with the way a person conducts himself or herself in the workplace



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	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

SELF-CHECK 3.1-1

- 1. Define workplace.
- 2. Define organization.
- 3. Create your own organizational chart using the following details:
 - Name of Farm
 - Location
 - Services
 - Tagline
 - Vision/Mission/ goal
- 4. Formulate your own code of ethics depends on the VMG of your farm.
- 5. List some procedures on how to maintain personal hygiene.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Answer Key 3.1.1

1. A place where people work, such as an office or factory ("health and safety in **the workplace**").
2. Organization An organization, or organization, is an entity – such as a company, an institution, or an association – comprising one or more people and having a particular purpose. The word is derived from the Greek word organon, which means tool or instrument, musical instrument, and organ. [Wikipedia](#)
3. JBuya Farm
La Trinidad, Benguet
Production and Learning Site
“All things are possible”
Vision: JBuya Farm envision for a community-based farm tourism, learning site and income generating farming system in the urban.
Mission: To provide safe, quality and fresh vegetable& strawberry with the use of innovative technology in production, care& Maintenance, Harvesting and pos-harvest handling to marketing; to encourage younger generations that farming is an opportunity and business; adoption of new technologies and enrichment of the soil through good agricultural practices.

4&5

code of ethics

A written set of guidelines issued by an organization to its workers and management to help them conduct their actions ...

BD BusinessDictionary

CODE OF ETHICS

1 CLIENT SERVICE

2 CONFLICT OF INTEREST

3 GROUP AUTONOMY

4 PROCESSES, METHODS AND TOOLS

5 RESPECT, SAFETY, EQUITY AND TRUST

6 STEWARDSHIP OF PROCESS

7 CONFIDENTIALITY

8 PROFESSIONAL DEVELOPMENT

6.

What is Personal Hygiene?

Regular Routine of Personal Care
Washing and Grooming of

- Hair
- Face and Skin
- Teeth
- Ears
- Hands
- Nails
- Feet



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:
			Page No.
			Revision:

TASK SHEET 3.1-1

Title: Construct your own Farm Organizational Chart		
Title Objective: At the end of the activity, you can be able to construct your own organizational chart		
Materials: Manila paper, scotch tape, pentel pen		
Procedure: Construct your own organizational chart following the above example on Sheet 3.1.1		
Self-Check	YES	NO
Did I		
Can construct organizational chart?		
Create own organizational chart?		
Must be displayed in the farm office?		
Maintain professional and personal growth		
Name of Trainee:		
Date:		
Evaluated by:		
Date		

JOB SHEET 3.1.1

Title: Construct your own vision, mission and goals		
Title Objective: At the end of the activity, you can be able to construct your own VMG		
Materials: Manila paper, scotch tape, pentel pen		
Procedure: Construct your own organizational chart following the above example on Sheet 3.1.1		
Self-Check	YES	NO
Did I		
Can I formulate my own VMG?		
Create own VMG based on your farms line of services?		
Must be displayed in the farm office?		
Maintain professional and personal growth		
Name of Trainee		
Date		
Evaluated by:		
Date		

SELF-ASSESSMENT GUIDE

Qualification	AGRICULTURAL CROP PRODUCTION NC II		
Certificate of Competency	PRACTICE CAREER PROFESSIONALISM		
Instruction			
•Read each of the questions.			
•Place check on each question to indicate your answers.			
Can I?		YES	NO

LO2. Set and meet work priorities

Assessment Criteria

- 1. Competing demands are prioritized to achieved personal, team and organizational goals and objective.
- 2. Resources are utilized efficiently and effectively to manage work priorities and commitment.
- 3. Practices along economic use and maintenance of equipment and facilities are followed as per established procedures.

Contents

- 1. Organizational Key Result Area (KRA).
- 2. Work values and ethical standards.
- 3. Company policies in the use and maintenance of equipment.

Conditions

- 1. Hands-out on Organizational KRA, Work Values and Ethics, Company policies and standards, and sample job targets.
- 2. Learning guides.
- 3. CD's

Methodologies

- 1. Interactive lecture
- 2. Demonstration
- 3. Structured activity
- 4. Group discussion

Assessment Methods

- 1. Role play
- 2. Interview
- 3. Written Assessment

SELF-CHECK
ANSWER KEY
TASK SHEET
JOB SHEET

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

Learning Experiences

Learning Outcome 2

LO2. Title: Set and meet work priorities

Learning Activities	Special Instructions
1. Read Information Sheet 2.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 2.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 1.1-1 and answer Self- Check 1.1-1 again.
3. Do Task Sheet 2.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 2.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

INFORMATION SHEET 2.1-1

Definition of terms

- 1.Goals
- 2. Work values
- 3. Ethical standards
- 4. Company policies



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

LO3. Maintain professional growth and development

Assessment Criteria

- 1. Trainings and career opportunities are identified an availed based on job requirements.
- 2. Recognition are soughs/received and demonstrated as proof of Career Advancement.
- 3. Licenses and certifications relevant to job and career are obtained and renewed.

Contents

- 1. Qualification standards
- 2. Gender and development sensitivity (GAD)
- 3. Professionalism in workplace
- 4. List of Professional Licenses

Condition

- 1. Quality standards
- 2. GAD Hand-outs
- 3. CD’s on Professionalism

Methodologies

- 1. Interactive lecture
- 2. Film viewing
- 3. Role play/ simulation
- 4. Group discussion

Assessment

- 1. Portfolio assessment
- 2. Interview
- 3. Simulation/Role play
- 4. Third Party Reports
- 5. Exams and quizzes

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

Learning Experiences

Learning Outcome 3

LO3. Title: Maintain professional growth and development

Learning Activities	Special Instructions
1. Read Information Sheet 3.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 3.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 1.1-1 and answer Self- Check 1.1-1 again.
3. Do Task Sheet 3.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 3.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	

INFORMATION SHEET 3.1-1

Sample Trainings:

- Leadership training
- Trainer’s Trainee

Sample of Recognitions for Career Advancement

- Awardee
- Resource speaker

Licenses or certifications



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

COMPETENCY-BASED LEARNING MATERIAL



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Issued by:	Page No.
			Revision:	



1

Hazardous Chemicals



2

Ladders



3

Scaffolding



4

Vehicle Accidents



5

Respiratory Hazards



6

Electrical Wiring



7

Machinery Malfunction



8

Falling Objects



9

Slippery / Wet Floors



10

Poor Housekeeping



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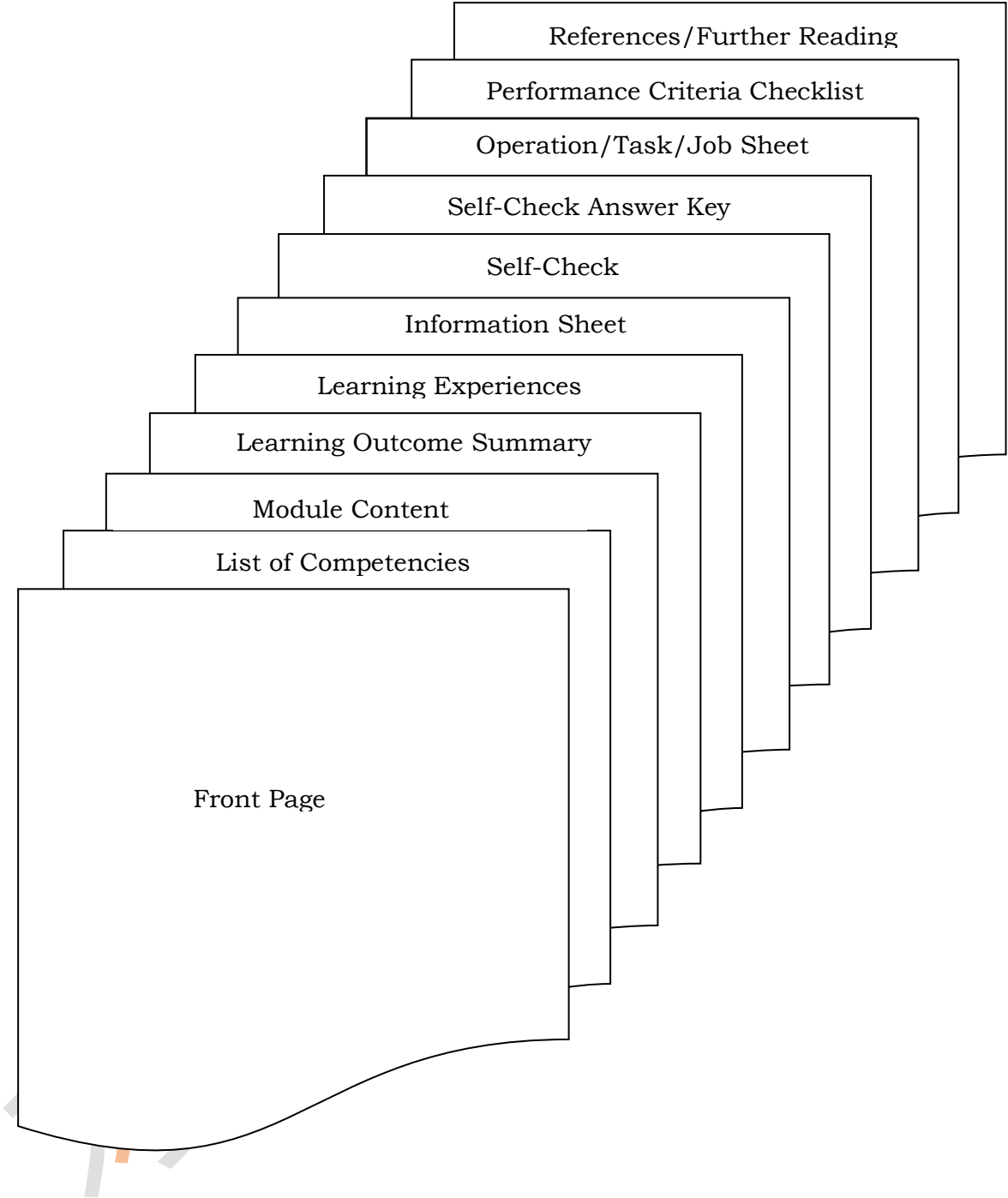
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Sector: Agriculture and Fishery
Qualification Title: Agricultural Crop production NC II
Unit of Competency: Practice Occupational Health and Safety Procedures
Module Title: Practicing Occupational Health and Safety Procedures

	AGRICULTURAL CROP PRODUCTION NC II	Date Developed: November 16, 2020	
		Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020 Issued by: Revision: Page No.

PARTS OF A COMPETENCY-BASED LEARNING MATERIAL



	AGRICULTURAL CROP PRODUCTION NC II	Date Developed:	Date Developed: November 16, 2020	
		November 16, 2020	Issued by:	Page No.
		Developed by: Jumelyn P. Buya		
			Revision:	

HOW TO USE THIS COMPETENCY-BASED LEARNING MATERIAL

Welcome!

The unit of competency, “Practice Occupational Health and Safety Procedures”, is one of the Common Competency of Agricultural Crop Production NC II, a course which comprises the knowledge, skills and attitudes required for National Certificate level II.

The module, Practice Occupational Health and Safety Procedures, contains training materials and activities related to enhancing the knowledge, skills and attitude of the trainees in agricultural crop for you to complete.

In this module, you are required to go through a series of learning activities in order to complete each learning outcome. In each learning outcome are Information Sheets, Self-Checks, Operation Sheets and Task/Job Sheets. Follow and perform the activities on your own. If you have questions, do not hesitate to ask for assistance from your facilitator.

Remember to:

- Work through all the information and complete the activities in each section.
- Read information sheets and complete the self-check. Suggested references are included to supplement the materials provided in this module.
- Most probably, your trainer will also be your supervisor or manager. He is there to support you and show you the correct way to do things.
- You will be given plenty of opportunities to ask questions and practice on the job. Make sure you practice your new skills during regular work shifts. This way, you will improve your speed, memory and your confidence.
- Use the Self-Checks, Operation Sheets or Task or Job Sheets at the end of each section to test your own progress. Use the Performance Criteria

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buaya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

Checklist or procedural Checklist located after the sheet to check your own performance.

- When you feel confident that you have had sufficient practice, ask your Trainer to evaluate you. The results of your assessment will be recorded in your Progress Chart and Accomplishment Chart.

You need to complete this module before you can perform the next module.

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

AGRICULTURAL CROP PRODUCTION NC II

List of Competencies

No.	Unit of Competency	Module Title	Code
1	Participate in workplace communication	Participating in workplace communication	
2	Work in a team	Working in a team	
3	Practice career professionalism	Practicing career professionalism	
4	Practice occupational health and safety procedure	Practice occupational health and safety procedure	

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

UNIT OF COMPETENCY: Practice Occupational Health and Safety Procedures

MODULE TITLE: Practicing Occupational Health and Safety Procedures

MODULE DESCRIPTOR:

This unit covers the outcomes required to comply with regulatory and organizational requirements for occupational health and safety

NOMINAL HOURS: 5 hours

SUMMARY OF LEARNING OUTCOMES:

LO1. Identify hazards and risks

LO2. Evaluate hazards and risk

LO3. Control hazard and risks

LO4. Maintain Occupational Health and Safety awareness (OHS)

DEAILS OF LEARNING OBJECTIVES

LO1. Identify hazards and risks

Assessment Criteria

1. Safety regulations and workplace safety-hazard control practices and procedures are clarified and explained based on organizational procedures.
2. Hazard/risks in the workplace and their corresponding indicators are identified to minimize or eliminate risks to co-workers, workplace and environment in accordance with organizational procedures.
3. Contingency measures during workplace accidents, fire and other emergencies are recognized and established in accordance with organizational procedures.

Contents

1. Hazards and risks identification and control
2. Organizational safety and health protocol
3. Contingency measures

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buaya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

4. Threshold limit value (TVL)
5. OHS indicators

Conditions

1. Workplace or assessment location
2. Occupational Health and Safety records
3. Personal and preventive
4. Equipment Health Records

Methodologies

1. Interactive-lecture
2. Film viewing
3. Role play/simulation
4. Group discussion

Assessment Methods

1. Portfolio Assessment
2. Interview
3. Case study/situation

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

Learning Experiences

Learning Outcome 1

LO1. Title: Identify hazards and risks

Learning Activities	Special Instructions
1. Read Information Sheet 1.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 1.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 1.1-1 and answer Self- Check 1.1-1 again.
3. Do Task Sheet 1.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 1.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

INFORMATION SHEET 1.1-1

Definition of terms

1. Hazards
2. Risks
3. Contingency measures
4. Threshold limit value (TVL)
5. OHS indicators
6. OHS
7. PPE



The Creation and Prevention of Hazards In The Workplace

Health and safety in the workplace depend on both the employer and the employees. There is no sole person responsible for the well-being of people who come into the workplace. Instead, there are a number of things that a number of people can do in order to create or contribute to a safe and healthy environment for all who enter.

There are some poor workplace practices that create hazards and put personnel at risk of harm or health problems. We'll show you some of the practices we're talking about.

- Unauthorized use of tools or machinery.
- Operating tools or machinery at unsafe speeds or in violation of the safety code of the workplace.
- Disablement or removal of safety guards or other such devices from machinery or equipment.
- The unsafe use or the use of defective tools or equipment.
- The use of body parts as opposed to the use of the necessary tools or push sticks.
- Improper use or handling of materials.
- The repair or adjustment of equipment that is currently in use and operates under pressure, with electricity, etc.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

- No use, misuse or non-maintenance of personal protective gear or safety equipment.
- Operating under suspended loads, scaffolds, open hatches, etc.
- Not practicing proper hygiene, lack of proper workplace maintenance, smoking in unauthorized areas and more.

You can prevent hazards in the workplace by having workplace inspections, which includes the following things:

- Paying attention or actively listening to the concerns of workers and supervisors in relation to health and safety concerns.
- Getting a better understanding of the various jobs and tasks at hand.
- Identifying potential and existing hazards.
- Determining the underlying causes for these hazards.
- Monitoring hazard controls (such as policies, procedures and safety equipment).
- Making recommendations on corrective actions.

A safe work place is crucial and ensuring that you have the right safety procedures and policies for things such as working at heights, first aid CPR, confined space entry, as well as things like WHMIS-GHS can create a great work environment that is safe and productive. Since Advanced Consulting and Training Ltd.'s founding, their diverse team of certified health and safety professionals have taken great pride in their ability to deliver prompt, cost-effective and relevant workplace health and safety solutions. As a MOL approved, TSSA accredited, and WSIB approved provider, we look forward to discussing how ACT can help with your company's safety requirements. Contact us today.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

LO2. Evaluate hazards and risk

Assessment Criteria

1. Terms of maximum tolerable limits which when exceeded will result in harm or damage are identified based on threshold limit values.
2. Occupational Health and Safety (OHS) issues and/or concerns and identified safety hazards are reported to designated personnel in accordance with workplace requirements and relevant workplace OHS legislations.

Contents

1. Safety Regulations
2. Hazards risks
3. Contingency measures
4. PPE
5. Emergency-related drills and trainings
6. Occupational Health and Safety personal records

Methodologies

1. Interactive lecture
2. Film viewing
3. Role plat/simulation
4. Group discussion

Assessment Criteria

1. Portfolio Assessment
2. Interview
3. Case study/situation

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

Learning Experiences

Learning Outcome 2

LO2. Title: Evaluate hazards and risk

Learning Activities	Special Instructions
1. Read Information Sheet 2.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 2.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 2.1-1 and answer Self- Check 2.1-1 again.
3. Do Task Sheet 2.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 2.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

LO3. Control hazard and risks

Assessment Criteria

1. Occupational Health and Safety (OHS) procedures for controlling hazards/risks in workplace are consistently followed.
2. Procedures for dealing with workplace accidents, fire and emergencies are followed in accordance with organization OHS policies.
3. Personal protective equipment (PPE) is correctly used in accordance with organization OHS procedures and practices.
4. Appropriate assistance is provided in the event of a workplace emergency in accordance with established organizational protocol.

Contents

1. Safety regulations
2. Clean air act
3. Electrical and fire safety code
4. Waste management
5. Disaster and preparedness and management
6. Contingency measures and procedures

Conditions

1. Hand-outs on-safety regulations, Electrical fire code, waste management disaster preparedness and management contingency measures and procedures.
2. Personal Protective Equipment (PPE)
3. OHS personal records control
4. CD's

Methodologies

1. Interactive-lecture
2. Film viewing
3. Role play/simulation
4. Group Discussion

Assessment methods

1. Demonstration
2. Interview
3. Case/situation analysis
4. Simulation

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

Learning Experiences

Learning Outcome 3

LO3. Title: Control hazard and risks

Learning Activities	Special Instructions
1. Read Information Sheet 3.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 3.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 3.1-1 and answer Self- Check 3.1-1 again.
3. Do Task Sheet 3.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 3.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

LO4. Maintain Occupational Health and Safety awareness (OHS)

Assessment Criteria

1. Emergency-related drills and trainings are participated in as per established organizational guidelines and procedures.
2. OHS personal records are completed and updated in accordance with workplace requirement.

Contents

1. Operational health and safety procedures and practices and regulations.
2. Emergency-related drills and trainings

Conditions

1. Workplace
2. Occupational Health and Safety records
3. Personal protective equipment
4. Health records

Methodologies

1. Interactive lecture
2. Simulation
3. Symposium
4. Group dynamics

Assessment Methods

1. Portfolio Assessment
2. Interview
3. Demonstration
4. Written Examination

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

INFORMATION SHEET

1. Occupational Health and Safety (OHS)
2. Occupational Health and Safety records
3. Workplace
4. Personal Protective Equipment
5. Clean Air Act Law
6. Solid Waste

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

Learning Experiences

Learning Outcome 4

LO4. Title: Maintain Occupational Health and Safety awareness (OHS)

Learning Activities	Special Instructions
1. Read Information Sheet 4.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 4.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 3.1-1 and answer Self- Check 3.1-1 again.
3. Do Task Sheet 4.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 4.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

UNIT OF COMPETENCY: Perform Estimation and Basic Calculation

MODULE TITLE: Performing Estimation and Basic Calculation

MODULE DESCRIPTOR

This module covers the knowledge, skills and attitudes required to perform estimation and basic workplace calculations

NOMINAL HOURS: 8 hours

SUMMARY OF LEARNING OUTCOMES:

LO1. Perform estimation

LO2. Perform basic workplace calculation

DETAILS OF LEARNING OUTCOMES

LO1. Perform estimation

Assessment Criteria

1. Job requirements are identified from written or oral communication.
2. Quantities of materials and resources required to complete a work task are estimated.
3. Time needed to complete a work is estimated.
4. Accurate estimate for work completion is made.
5. Estimate of materials and resources are reported to appropriate person.

Contents

1. Problem solving procedures
2. Basic mathematical operations

Conditions

1. Pen/pencil
2. Paper
3. Sample problems

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

Methodologies

1. Computation
2. Estimation
3. Interaction

Methods of Assessment

1. Oral questioning
2. Interview

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

INFORMATION SHEET 3.1-1

- Fertilizer Calculation
- Farm Plan
- Cash Statement
- Cash book

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

Learning Experiences

Learning Outcome 1

LO1. Title: Perform estimation

Learning Activities	Special Instructions
1. Read Information Sheet 3.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 3.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 1.1-1 and answer Self- Check 1.1-1 again.
3. Do Task Sheet 3.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

LO2. Perform basic workplace calculation

Assessment Criteria

1. Calculation to be made are identified according to job requirement.
2. Correct method of calculation is determined.
3. Systems and unit of measurement to be followed are ascertained.
4. Calculations needed to complete work task are performed using the four basic mathematical operations.
5. Appropriate operations are used to comply with the instructions.
6. Result obtained is reviewed and thoroughly checked.

Contents

1. Basic mathematical operations
2. Systems of measurements
3. Units of measurements
4. Conversion of units
5. Fractions and decimals
6. Percentages and ratios
7. Basic accounting principles and procedures

Conditions

1. Engine pump
2. Calculator
3. Paper
4. Reference material
5. Sample problems/worksheets
6. Conversion table

Methodologies

1. Self paced modular
2. Lecture/discussion
3. Interaction
4. Practical exercises
5. Computation/worksheet

Assessment Methods

1. Oral/written examination
2. Practical exercises
3. Practical demonstration

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

INFORMATION SHEET 3.1-1

- Basic mathematical symbols
- Units of measurements
- Conversion of units
- Percentage germination

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

Learning Experiences

Learning Outcome 2

LO2. Title: Perform basic workplace calculations

Learning Activities	Special Instructions
1. Read Information Sheet 2.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
2. Answer Self Check 2.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 2.1-1 and answer Self- Check 2.1-1 again.
3. Do Task Sheet 2.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
4. Do Task Sheet 2.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

UNIT OF COMPETENCY: Process Farm Wastes

MODULE TITLE: Processing farm wastes

MODULE DESCRIPTOR:

This module covers the knowledge, skills, and attitudes required in proper handling of farm wastes. It comprises functions such as collecting farm wastes, identification, segregation and processing of farm wastes as well as performing the house keeping duties.

NOMINAL HOURS: 12 hours

SUMMARY OF LEARNING OUTCOMES:

- LO1. Collect farm wastes
- LO2. Identify and segregate wastes
- LO3. Treat and process wastes
- LO4. Perform housekeeping

DETAILS OF LEARNING OUTCOMES:

LO1. Collect farm wastes

Assessment Criteria

1. Tools and materials are prepared for collection of farm wastes.
2. Wastes are collected following OSHS and waste collection requirements and plan.
3. Appropriate PPE worn as prescribed by OSHS

Contents

1. Knowledge, theory practices and systems operations
2. Tools and materials used in waste management
3. What is 5's
4. Safety practices
5. Personal protective equipment
6. MSDS mathematics and mensuration
7. Profitability of the operation
8. Volume of wastes
9. Occupational health and safety

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

10. Skills in using tools and equipment values
11. Diligence
12. Time consciousness
13. Personal integrity in doing routine management practices
14. Perseverance in executing routine works
15. Ability to work with others harmoniously

Conditions

1. Tools

- Spade
- wheel borrow
- broomstick
- sprayer or pressurized pump

2. Materials

- goggles
- disposal gloves
- face mask
- rubber boots
- overall
- first aid kit
- chemical spill kit

Methodologies

1. Lecture/discussion
2. Practical demonstration

Assessment Method

1. Written Test
2. Observation and questioning
3. Demonstration

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

INFORMATION SHEET 1.1-1

Composting-

Tools and equipment used in composting

- Wheel borrow
- Shredder
- Bolo
- Sickles
- Chopping board
- Shovel

Materials used in composting

- Green materials (source of nitrogen)
- Brown materials (source of carbon)
- Kitchen wastes
- Vegetable trimmings
- Chicken manure
- Top soil
- Black plastic mulch
- PVC pipe
- Activator

Factors affecting composting

- Materials
- Temperature
- Climate
- Drainage
- Microorganisms

Benefits of composting

1. Improves soil structure.
2. Safe and easy to use.
3. Cleans the environment.
4. Increase organic matter content of the soil.
5. Improves chemical and biological properties of the soil.
6. Increase water holding capacity
7. Economical and cheap

Procedure of composting:

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buaya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

1. Site selection. Good drainage condition.
2. Gather all materials and chopped or shred into small pieces/cut for easier decomposition process.
3. Pile the brown materials followed by green materials at 1:1:1 (C:N ratio) part of pile, then chicken manure/pig/cow manure. Maintain moisture and cover with soil. Repeat procedure until pile is about 5-6 ft in height.
4. Place the PVC at the middle of the pile in order to maintain aeration and temperature.
5. Spray with activator to hasten decomposition process.
6. Maintain moisture.
7. Cover with black plastic mulch to maintain temperature.
8. Check temperature.
9. After 3 weeks, turn your pile.
10. Every week check temperature and moisture and turn your pile until ready for use.

Characteristic of compost

1. Black in color
2. Odorless

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

Learning Experiences

Learning Outcome 2

LO2. Title: Identify and segregate wastes

Learning Activities	Special Instructions
5. Read Information Sheet 2.1-1 on the integrate personal objectives with organizational goals	If you believe you understood the Information Sheet, proceed to the next. If not, read the Information Sheet again.
6. Answer Self Check 2.1-2.	If you get a perfect score, proceed to the next activity. If not, read Information Sheet 2.1-1 and answer Self- Check 2.1-1 again.
7. Do Task Sheet 2.1-3	If you got a satisfactory rating, proceed to the next. If not, read and do the Task Sheet again.
8. Do Task Sheet 2.1-4	If you got a satisfactory rating proceed to the next. If not, read and do the Task Sheet again.

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

LO3. Treat and process wastes

Assessment criteria

1. Processing of wastes generated at the farm is done according by categories following environmental legislation and codes RA 9003 and RA 6969.
2. Principles of 3R's (reduce, recycle& re-used) are applied accordingly
3. Wastes generated at the farm disposed according to environmental legislation and codes RA 9003 and RA 6969.
4. Dangerous and hazardous wastes are handled according to organizational requirements and relevant legislation following OSHS procedures.

Contents

1. Knowledge, theory, practices and system operations principles of 3R's
2. Dangerous and hazardous wastes
3. Handling, storage and disposal procedures
4. Systems, processes and operation
5. Processing of wastes
6. Standard procedures in handling dangerous and hazardous wastes
7. Codes and regulations
8. Appropriate legal regulatory body that supervises the animal industry such BAI and Philippine Animal Health Center (PAHC), DENR, DOH, BFAD
9. Occupational safety and health standards
10. RA 6969

Conditions

1. Farm site
2. Instructional materials: Reference books, Professional emergency, learning guides, Modules, CD's, VHS tapes, transparencies- (now powerpoint presentations)
3. Different farm wastes
4. Tools and materials in waste management
5. Pencil/pen
6. Paper
7. Sample problem

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buaya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

Methodologies

1. Lecture/discussion-powerpoint presentations
2. Practical demonstration-actual hands-on

Assessment Methods

1. Written examination
2. Observation and questioning
3. Demonstration (practicum)

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

INFORMATION SHEET 4.1-1

- RA 9003
- RA 6969
- 3R's
- Codes and regulations
- Occupational safety and health standards

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

LO4. Perform housekeeping

Assessment Criteria

1. Work area is cleaned according to industry standard procedures.
2. Tools are checked, cleaned and stored according to established industry procedures and following user's manual.
3. Materials and other consumables are stored following industry standard procedures and manufacturer's specifications.
4. PPE is checked for damage prior to ensuring that clean and undamaged equipment is stored.
5. Storage facility is checked to ensure contamination in the area according to organizational requirements and legislation and codes.
6. Record keeping was done according to industry requirements.

Contents

1. Knowledge, theory, practices and systems operations
2. Tools and materials used in wastes management
3. Categories of wastes
4. Dangerous and hazard wastes
5. Handling, storage and disposal procedures
6. 5's safety practices
7. Personal protective equipment (PPE)
8. MSDS materials, uses and specifications
9. Office supplies and equipment- used for proper monitoring of production activities
10. Communications
11. Preparation of inventory reports and production records maintenance
12. Diligence
13. Personal integrity in doing routine management practices
14. Perseverance in executing routine works
15. Ability to work with others harmoniously

Conditions

1. Tools
 - Spade

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

- Wheel borrow
- Broomstick
- Sprayer or pressurized pump

2. Materials

- Sacks
- Containers
- Disinfectants
- Pen/pencil
- Paper

3. PPE

- Sacks
- Disposal gloves
- Face mask
- Rubber boots
- Overall

4. OHS

- First-aid kit
- Chemical spill kit

Methodologies

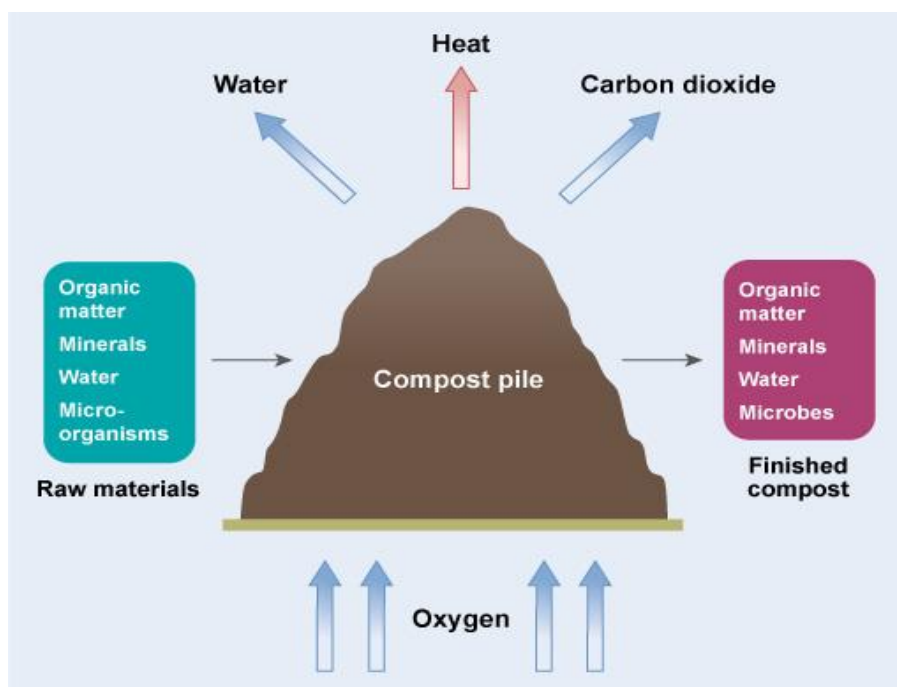
1. Lecture/ discussion
2. Practical demonstration

Assessment Methods

1. Written examination
2. Observation and questioning
3. Demonstration

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	

Composting is the process where biodegradable organic wastes (food and garden waste) are converted into compost in a natural biological process. Composting can be done by individual householders and community groups or on a commercial scale. On the larger scale, the waste from an entire town or city could be composted if sufficient land, labour and equipment is available. The benefits of composting are not only the reduction of waste, but also the production of compost which is a valuable **soil improver**. Soils treated with compost are better able to withstand droughts and are more fertile because plant nutrients are returned to the soil, which reduces the need for manufactured fertilizers. It is possible to add a certain amount of animal manure to residential waste for composting, which may help with other waste problems in the community and adds to the amount of useful soil improver that is made.



	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of ____
			Revision:	



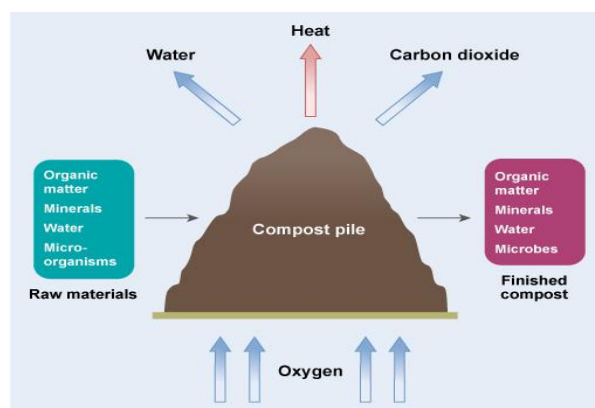
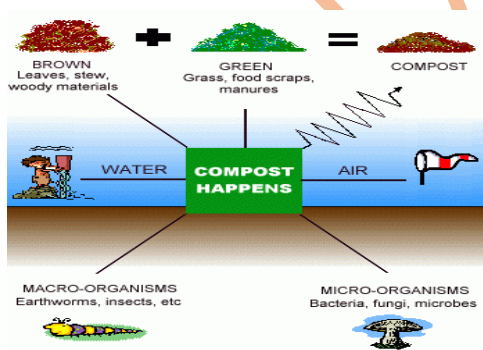
Collection of materials and shredding into small sizes



Compost pile



turning compost pile



AGRICULTURAL
CROP PRODUCTION
NC II

Developed by:
Jumelyn P. Buya

Date Developed:
November 16, 2020

Issued
by:

Page No. 13 of ____

Revision:

JOB SHEET 2.1-1

Title: Process farm waste		
Objective: 1. The trainee must be able to familiarize the materials, tools and equipment of processing farm waste. 2. To know the types of composting. 3. To be able to know the benefits of composting. 4. Enumerate the factors in composting. 5. Do actual compost piling. 6. Perform the 7's in basic housekeeping.		
Materials: brown materials (rice hulls, dried leaves, rice bran, coco dust) Green materials (sunflower leaves, all vegetable & fruit trimmings) Manure Tools: bolo, chopping board, black plastic mulch, shovel Equipment: shredder		
Procedure: 1. Collect all the needed materials. 2. Select well-drained area with shade for compost piling 3. Chopped all materials into small pieces ready for piling. 4. Measure a 4mx4m height and width. 5. Pile first the brown materials with equal measurement followed by green materials and animal manure. Maintain moisture. Cover with soil. 6. repeat the steps until the compost pile. 7. Cover the compost with black plastic much. 8. Maintain aeration by not compacting the pile. Use bamboo stick or PVC pipe to serve as exhaust in the center of the pile.		
Can I?	YES	NO
1. Identify the materials, tools and equipment?		
2. Collect materials for composting?		
3. Can operate shredder?		
4. Can do compost piling according to process?		
5. Perform basic housekeeping?		
Name of Trainee: _____ Date of performance: _____		

	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buaya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	

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	AGRICULTURAL CROP PRODUCTION NC II	Developed by: Jumelyn P. Buya	Date Developed: November 16, 2020	
			Issued by:	Page No. 13 of _____
			Revision:	